



RoseMirrorHA

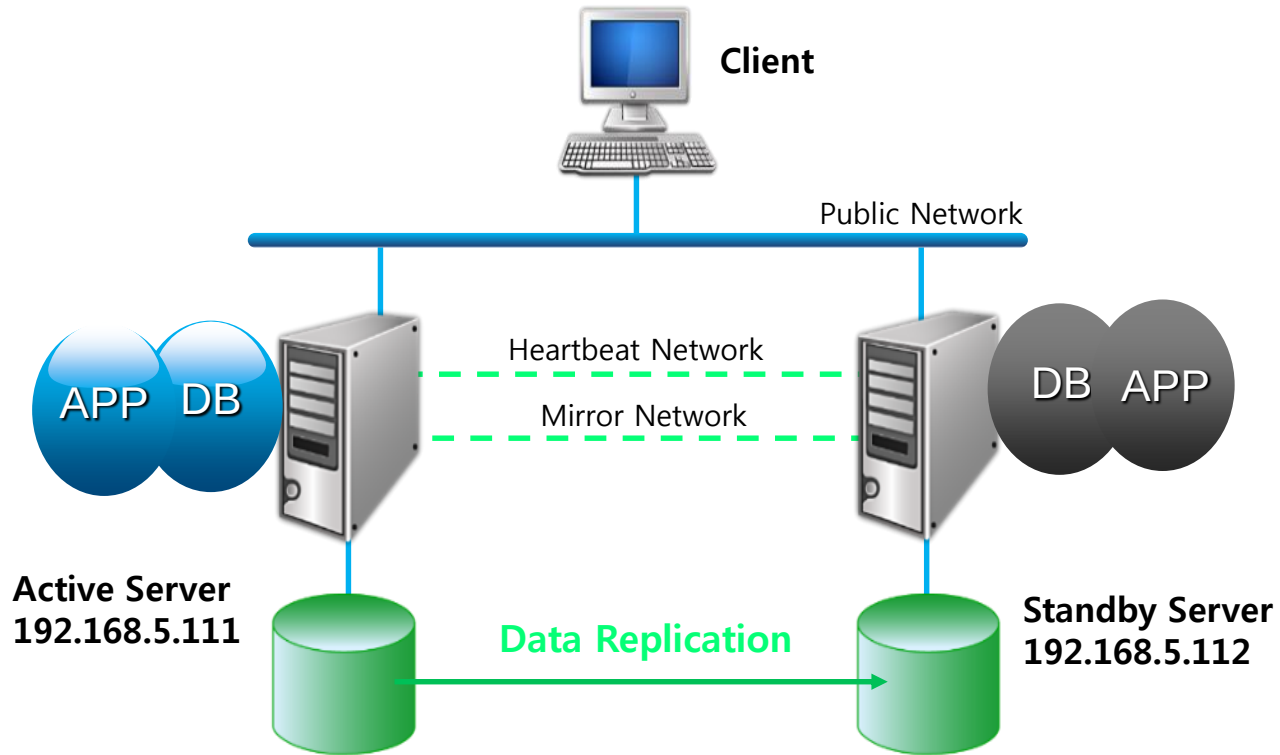
운영가이드

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1. RoseMirrorHA 구성도

1. RoseMirrorHA 구성도

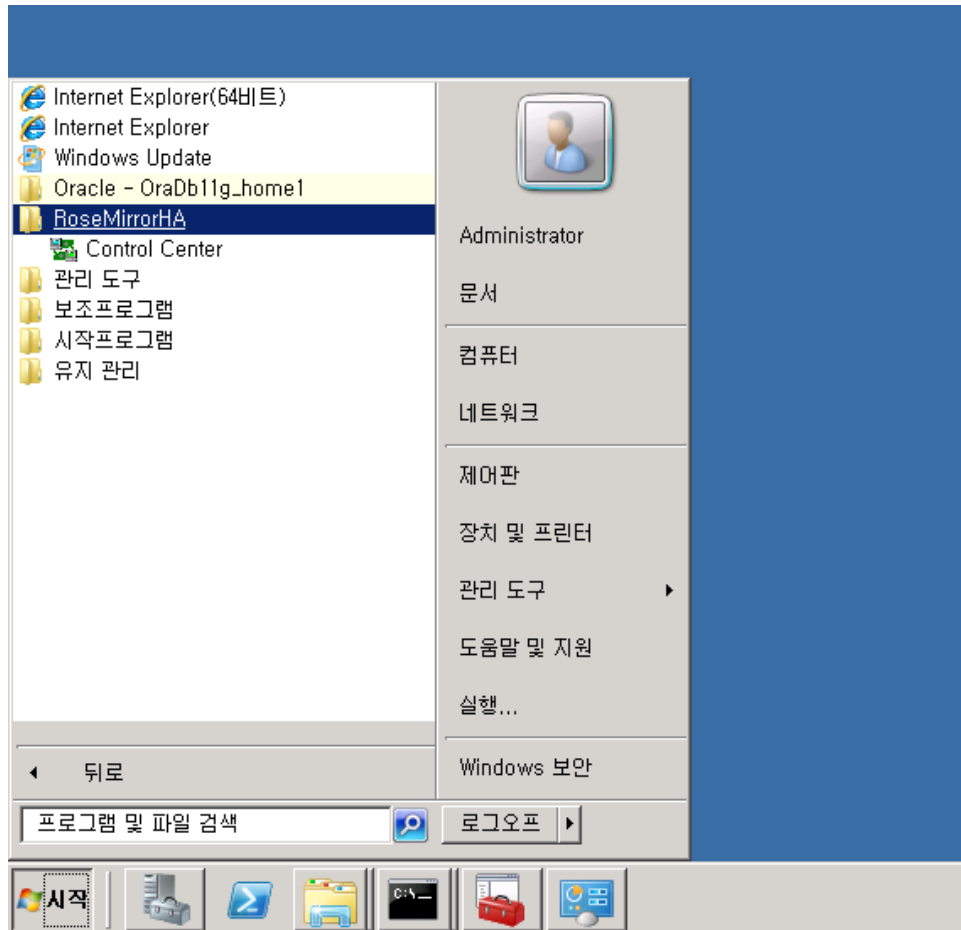


2. RoseMirrorHA Console

2. RoseMirroHA Console

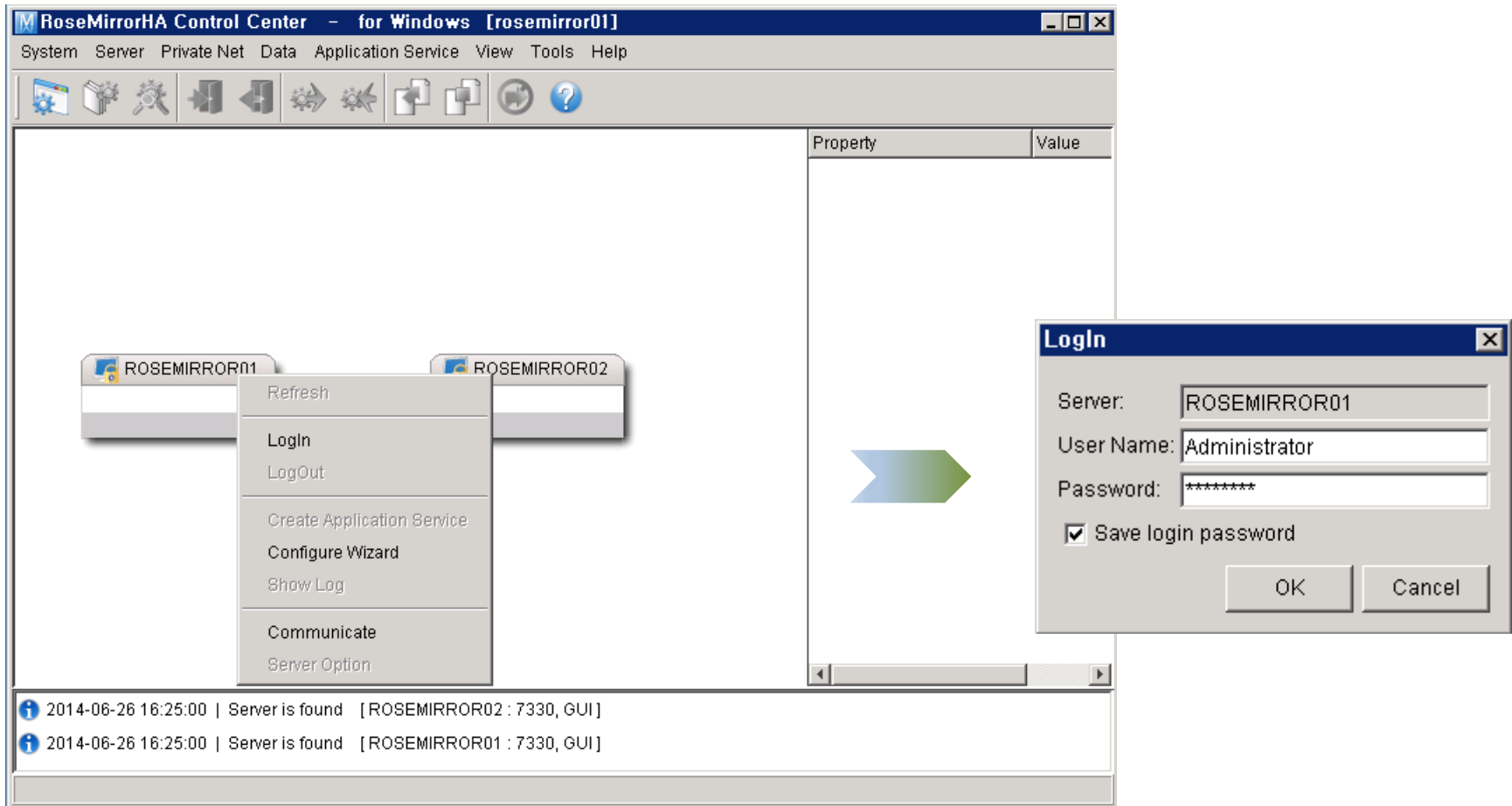
◆ RoseMirrorHA Console : 시작

- Windows 시작 >> 모든 프로그램 >> RoseMirrorHA >> Control Center



2. RoseMirroHA Console

- ◆ RoseMirror01(Hostname) 오른쪽 마우스 클릭 >> Login 선택
- ◆ Username / Password 입력 후 OK 선택 (ex, Administrator / P@ssw0rd)



2. RoseMirroHA Console

◆ RoseMirrorHA Console 설명(1)

The screenshot displays the RoseMirrorHA Control Center interface. On the left, a diagram shows two server nodes, ROSEMIRROR01 and ROSEMIRROR02, each containing an 'ORA_Group' component, connected by a line. A red rectangle highlights this diagram area. On the right, a 'Property' table lists various system parameters and their current values. A green arrow points from the 'Application service' property to the list of tasks on the far right.

Property	Value
Data	ORA_Group
Connection	Stop
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	0
Mirror	Stop
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Stop
Sent Bytes	1,510,111,472
Queued Bytes	0
Mode	Async
Target Side	Stop
Pagefile Used	0
Application service	ORA_Group

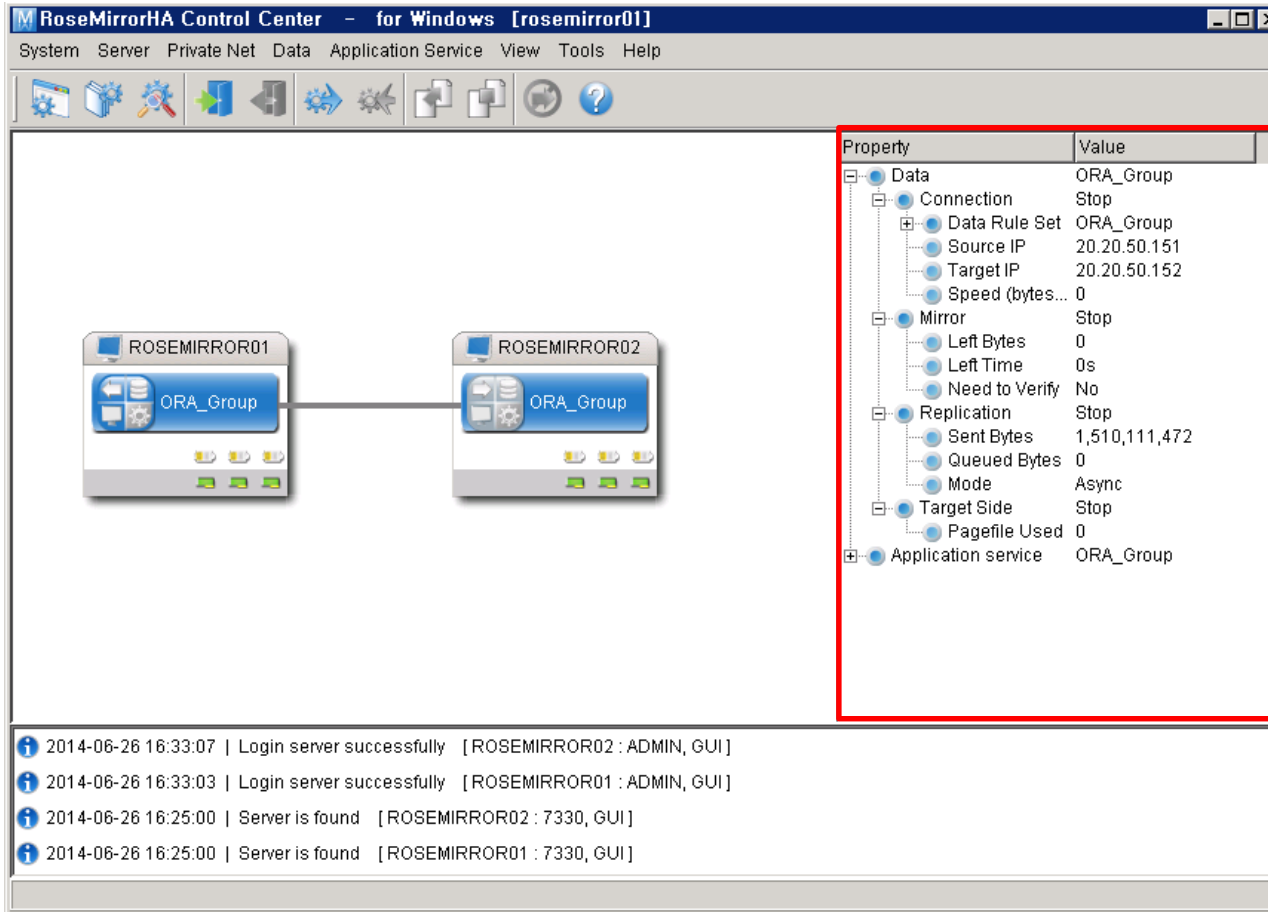
Log messages at the bottom:

- 2014-06-26 16:33:07 | Login server successfully [ROSEMIRROR02 : ADMIN, GUI]
- 2014-06-26 16:33:03 | Login server successfully [ROSEMIRROR01 : ADMIN, GUI]
- 2014-06-26 16:25:00 | Server is found [ROSEMIRROR02 : 7330, GUI]
- 2014-06-26 16:25:00 | Server is found [ROSEMIRROR01 : 7330, GUI]

- Server 상태 확인
- Group 상태 확인
- Agent 통신 확인
- NICs 상태 확인

2. RoseMirroHA Console

◆ RoseMirrorHA Console 설명(2)



The screenshot displays the RoseMirrorHA Control Center interface. On the left, two server icons labeled ROSEMIRROR01 and ROSEMIRROR02 are connected by a line, each showing the ORA_Group service. On the right, a 'Property' table is highlighted with a red border. A green arrow points from this table to a list of actions on the right side of the image.

Property	Value
Data	ORA_Group
Connection	Stop
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...)	0
Mirror	Stop
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Stop
Sent Bytes	1,510,111,472
Queued Bytes	0
Mode	Async
Target Side	Stop
Pagefile Used	0
Application service	ORA_Group

- Server 정보 확인
- Service 정보 확인

2. RoseMirroHA Console

◆ RoseMirrorHA Console 설명(3)

The screenshot displays the RoseMirrorHA Control Center interface. On the left, two server icons labeled ROSEMIRROR01 and ROSEMIRROR02 are connected by a line, each containing an 'ORA_Group' component. The right pane shows a tree view of properties for the selected 'ORA_Group'.

Property	Value
Data	ORA_Group
Connection	Stop
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...)	0
Mirror	Stop
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Stop
Sent Bytes	1,510,111,472
Queued Bytes	0
Mode	Async
Target Side	Stop
Pagefile Used	0
Application service	ORA_Group

At the bottom, a log window shows the following messages:

- 2014-06-26 16:33:07 | Login server successfully [ROSEMIRROR02 : ADMIN, GUI]
- 2014-06-26 16:33:03 | Login server successfully [ROSEMIRROR01 : ADMIN, GUI]
- 2014-06-26 16:25:00 | Server is found [ROSEMIRROR02 : 7330, GUI]
- 2014-06-26 16:25:00 | Server is found [ROSEMIRROR01 : 7330, GUI]

• Log 확인

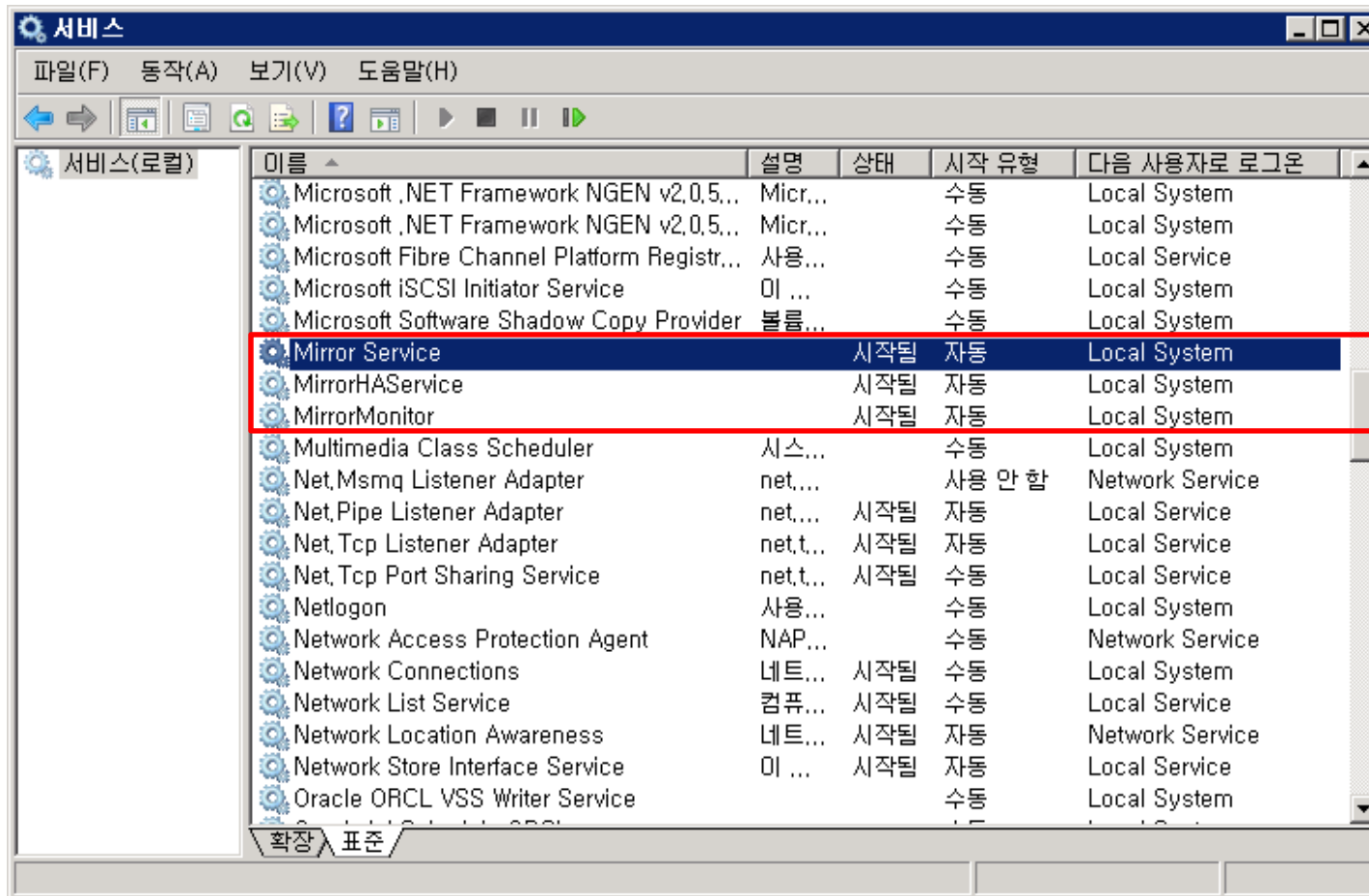
3. RoseMirrorHA Application Service and Group

- RoseMirrorHA Service
- RoseMirrorHA Process
- RoseMirrorHA Resource
 - Virtual IP
 - Data
 - NTService
 - File share
 - Agent
- RoseMirrorHA Group

3. RoseMirroHA Application Service and Group

◆ RoseMirrorHA : Service

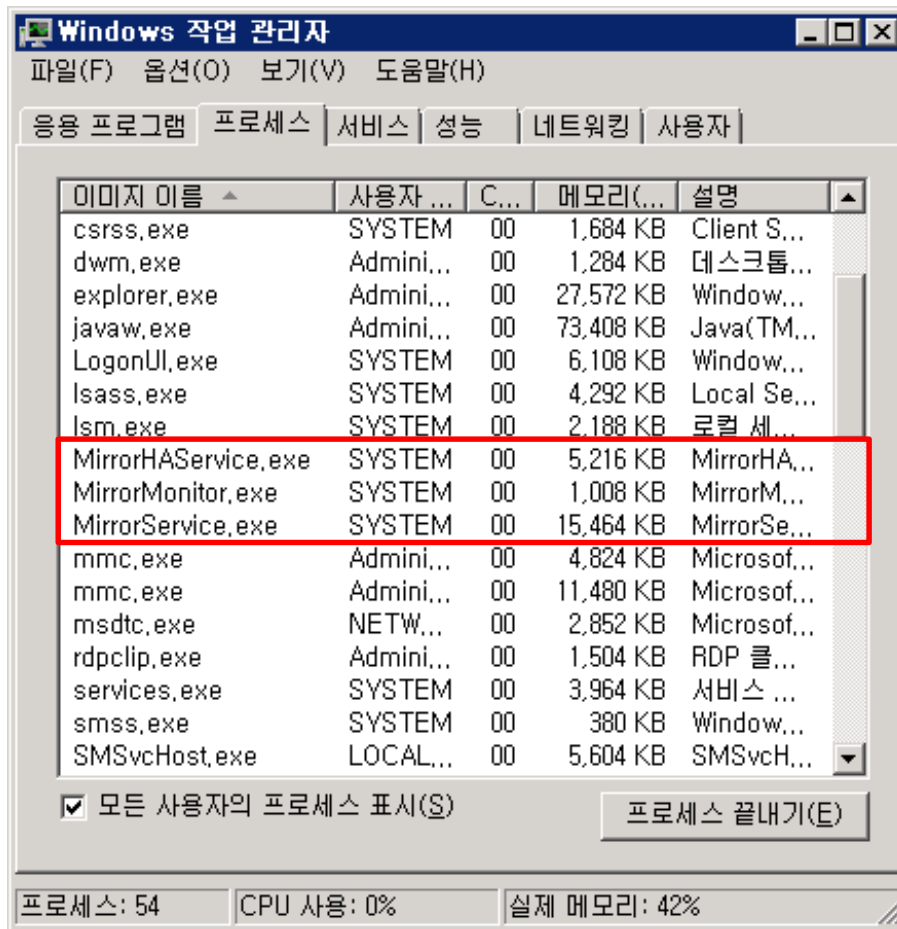
- Mirror Service, MirrorHAService, MirrorMonitor



3. RoseMirroHA Application Service and Group

◆ RoseMirrorHA : Process

- MirrorService.exe, MirrorHAService.exe, MirrorMonitor.exe



3. RoseMirroHA Application Service and Group

◆ RoseMirrorHA Resources

• Virtual IP

- Active IP : Service IP로 Client로부터 Service를 위해 연결하는 IP지정
- Alias : Service Name으로 Hostname처럼 사용할 수 있는 Name 지정
- Virtual Mac Address : Service IP가 할당되면 NIC의 Mac이 아닌 지정된 가상의 Mac Address가 지정
- Replace IP : Service IP를 할당하면, 현재의 Public IP를 대체하여 지정

Modify

Active IP

IP	SubNet Prefix
192.168.5.153	255.255.255.0

☐ Virtual Mac Address 0C-0b-09-0c-05-0d

☐ Replace IP

☒ Alias ORA

Holding NIC

ROSEMIRROR01	ROSEMIRROR02
☒ public (192.168.5.151;)	☒ public (192.168.5.152;)
☐ heartbeat (10.10.50.151;)	☐ heartbeat (10.10.50.152;)
☐ mirror (20.20.50.151;)	☐ mirror (20.20.50.152;)

OK Cancel

3. RoseMirroHA Application Service and Group

◆ RoseMirrorHA Resources

- Data
 - Replication Data : Volume, Folder, File를 Active to Standby node로 복제

The screenshot shows the 'Modify Application Service' dialog box with the 'Data' tab selected. The 'Bind data' checkbox is checked. Under 'Select IP address for replicating', ROSEMIRROR01 is set to 20.20.50.151 mirror and ROSEMIRROR02 is set to 20.20.50.152 mirror. Under 'Select replication model of data', the 'Async' radio button is selected, with 'Build Packet Timeout' set to 50 milliseconds. A table lists the rule path 'D:\oradata\'. The 'Include' and 'Recurse' checkboxes are checked. The 'Use Data Lock in Standby Server' checkbox is also checked. The 'Modify' button is visible on the right side of the table area. At the bottom are 'OK' and 'Cancel' buttons.

Rule Path	Rule Attributes
D:\oradata\	☒ Include ☒ Recurse

3. RoseMirroHA Application Service and Group

◆ RoseMirrorHA Resources

- Data
 - Share Disk : san, iSCSI 등의 방식으로 share disk를 구성하여 사용
 - Arbitral Disk : Split Brain 방지를 위해 Group의 소유권을 확인하기 위한 Volume를 등록

The screenshot shows the 'Modify Application Service' dialog box with the 'Share Disk' tab selected. The 'Shutdown takeover wait time' is set to 'Default' seconds. Below this, there are two columns for 'Mount Point', 'ROSEMIRROR01', and 'ROSEMIRROR02'. A 'Shared Disk' sub-dialog is open, displaying a table with columns: Server, Disk, Partion count, and Disk ID. The table contains two rows: ROSEMIRRO... Disk1 with partion count 1 and disk ID ced3d11321f0f86f3310bc4bd7ce58d, and ROSEMIRRO... Disk1 with partion count 1 and disk ID 517193bc2b2100f375894ee705542d6. Below the table are 'Disk parameter' and 'Volume parameter' sections. 'Disk parameter' includes 'Use hardlock' (SCSI2), 'Check timeout' (120 S), and checkboxes for 'Auto match disk number' (unchecked) and 'Standby monitor' (checked). 'Volume parameter' includes 'FS check' (Auto) and 'Mount point' (D). 'OK' and 'Cancel' buttons are at the bottom.

Server	Disk	Partion count	Disk ID
ROSEMIRRO...	Disk1	1	ced3d11321f0f86f3310bc4bd7ce58d
ROSEMIRRO...	Disk1	1	517193bc2b2100f375894ee705542d6

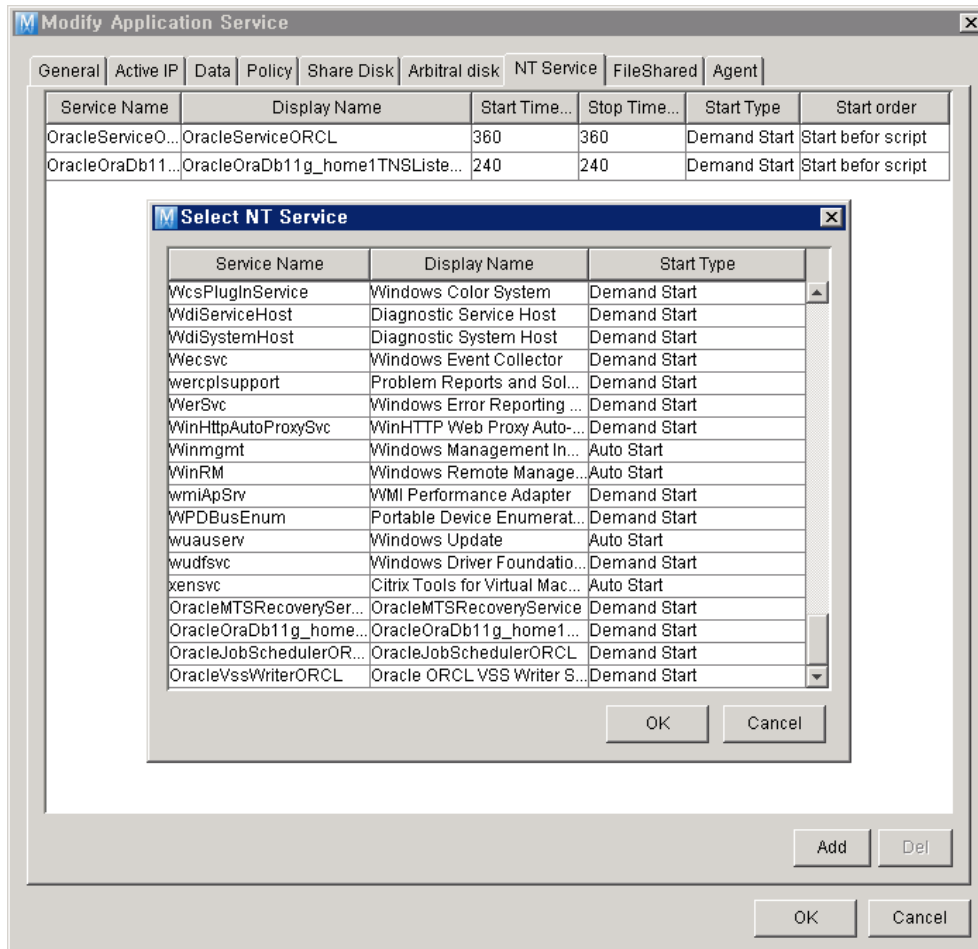
The screenshot shows the 'Modify Application Service' dialog box with the 'Arbitral disk' tab selected. The 'Check timeout' is set to 10 seconds. Below this, there are two columns for 'Disk Group', 'ROSEMIRROR01', and 'ROSEMIRROR02'. An 'Arbitral disk' sub-dialog is open, showing a 'Select exist disk' checkbox (checked) and a dropdown menu. Below this is the 'Arbitral disk group name:' field. A 'Disk' table is displayed with columns: Server, Disk, and Disk id. The table contains two rows: ROSEMIRROR01 and ROSEMIRROR02. Below the table is the 'Auto match disk number' checkbox (unchecked). 'OK' and 'Cancel' buttons are at the bottom.

Server	Disk	Disk id
ROSEMIRROR01		
ROSEMIRROR02		

3. RoseMirroHA Application Service and Group

◆ RoseMirrorHA Resources

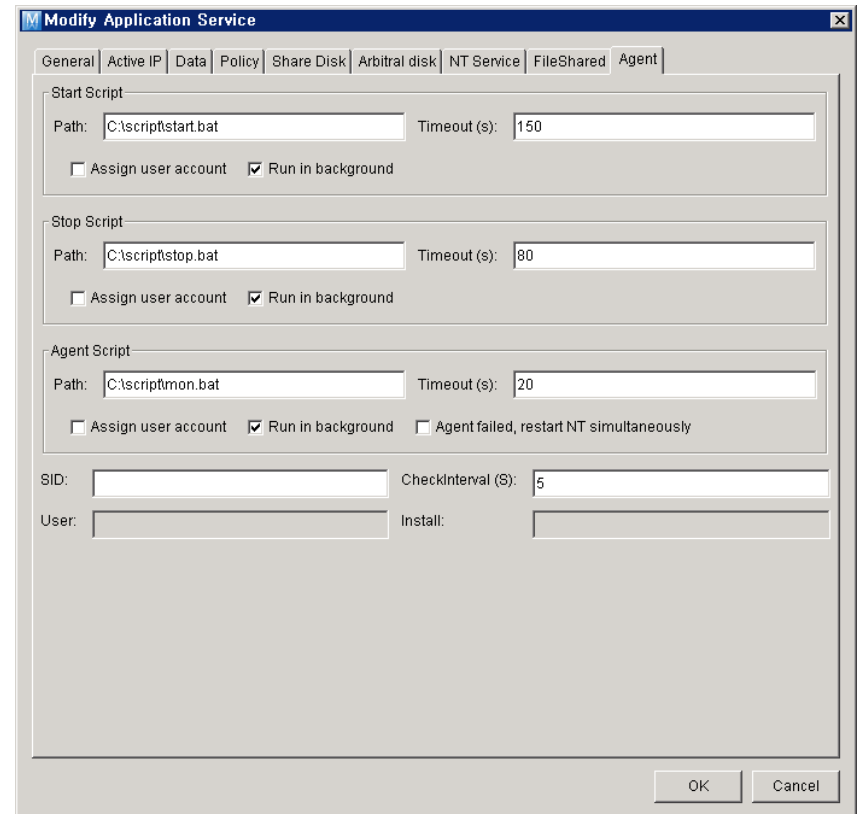
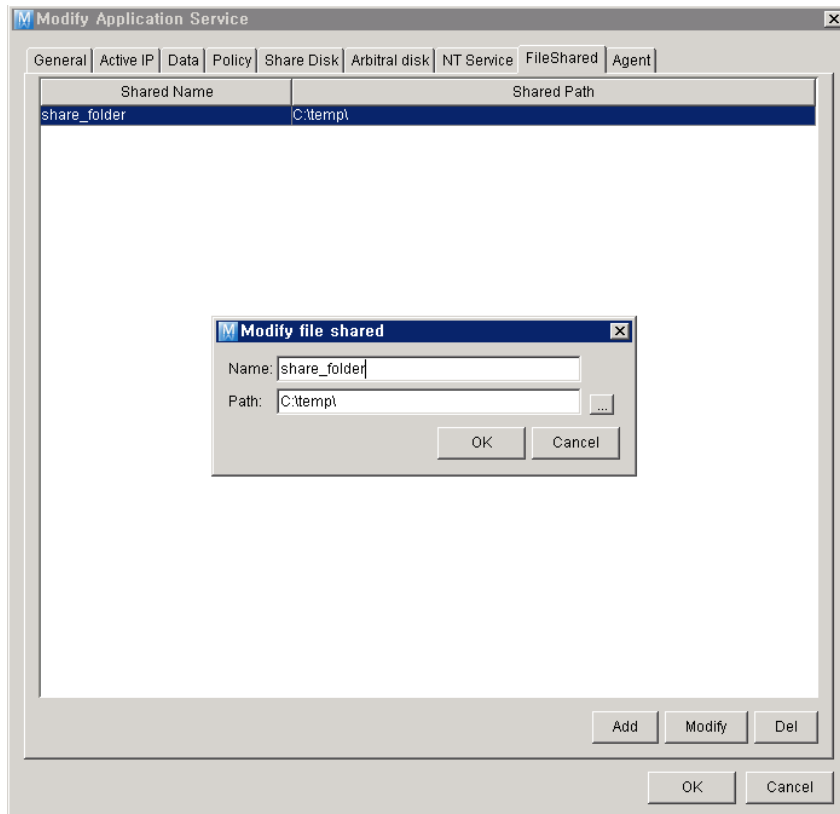
- NT Service : Windows Server에서 제공하는 Service를 등록



3. RoseMirroHA Application Service and Group

◆ RoseMirrorHA Resources

- File Shared : 공유폴더를 등록하여, Standby node로 절체되어도 동일한 공유폴더를 제공하기 위한 리소스
- Agent : Process 등록 및 별도의 Monitoring을 위해 사용



3. RoseMirroHA Application Service and Group

◆ RoseMirrorHA Group

- Virtual IP, Data, NT Service, File Share, Agent를 Group화 하여, 서비스 제공



Property	Value
Data	ORA_Group
Connection	Stop
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	0
Mirror	Stop
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Stop
Sent Bytes	1,510,111,472
Queued Bytes	0
Mode	Async
Target Side	Stop
Pagefile Used	0
Application service	ORA_Group
General	
Active IP	
NT Service	
Host Alias	

4. RoseMirrorHA 그룹 조작

- Group 실행하기 (Bring In)
- Group 이동하기 (Fail Over or Take Over)
- Group 중지하기 (Bring Out)
- Group 강제 실행하기 (Force start)
- Group 삭제하기 (Delete)

4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : 실행하기 (Bring In)

- Group에 등록된 Resources를 Online
- Group에서 오른쪽 마우스 클릭 후 "Bring In"선택
- Active - Standby node 중 서비스를 제공하지 않은 상태에서 가능

The screenshot displays the RoseMirrorHA Control Center for Windows. The main window shows two nodes, ROSEMIRROR01 and ROSEMIRROR02, both associated with the 'ORA_Group'. A context menu is open over the ROSEMIRROR01 node, with 'Bring In' highlighted. The right pane shows the properties of the selected node, including Data, Connection, Data Rule Set, Mirror, Replication, Target Side, and Application service. The bottom pane shows a log of recent events.

Property	Value
Data	ORA_Group
Connection	Stop
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	0
Mirror	Stop
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Stop
Sent Bytes	1,510,111,472
Queued Bytes	0
Mode	Async
Target Side	Stop
Pagefile Used	0
Application service	ORA_Group
General	
Active IP	
NT Service	
Host Alias	

Log entries:

- 2014-06-26 16:33:08 | Client: ROSEMIRROR02 | Successfully, client type : (rcc), login level: admin. [ROSEMIRROR02]
- 2014-06-26 16:33:04 | Client: ROSEMIRROR01 | Successfully, client type : (rcc), login level: admin. [ROSEMIRROR01]
- 2014-06-26 15:21:37 | The pri [ROSEMIRROR01]
- 2014-06-26 15:21:36 | The pri [ROSEMIRROR02]
- 2014-06-26 15:21:33 | Remote [ROSEMIRROR02]
- 2014-06-26 15:21:33 | The NI ROSEMIRROR01]

4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : 실행하기 (Bring In)

- Group이 정상 Online된 후의 화면

The screenshot displays the RoseMirrorHA Control Center for Windows. The main workspace shows two nodes, ROSEMIRROR01 and ROSEMIRROR02, each with an 'ORA_Group' component. A green line connects the two nodes, indicating a successful connection. The right-hand pane shows the properties for the selected 'ORA_Group' on ROSEMIRROR02, including Data, Connection, Data Rule Set, Mirror, Replication, and Application service. The bottom log window shows a series of messages indicating the successful completion of synchronization and connection verification.

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...)	0
Mirror	Idle
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Idle
Sent Bytes	1,512,709,248
Queued Bytes	0
Mode	Async
Target Side	I/O Busy
Pagefile Used	0
Application service	ORA_Group
General	
Active IP	
NT Service	
Host Alias	

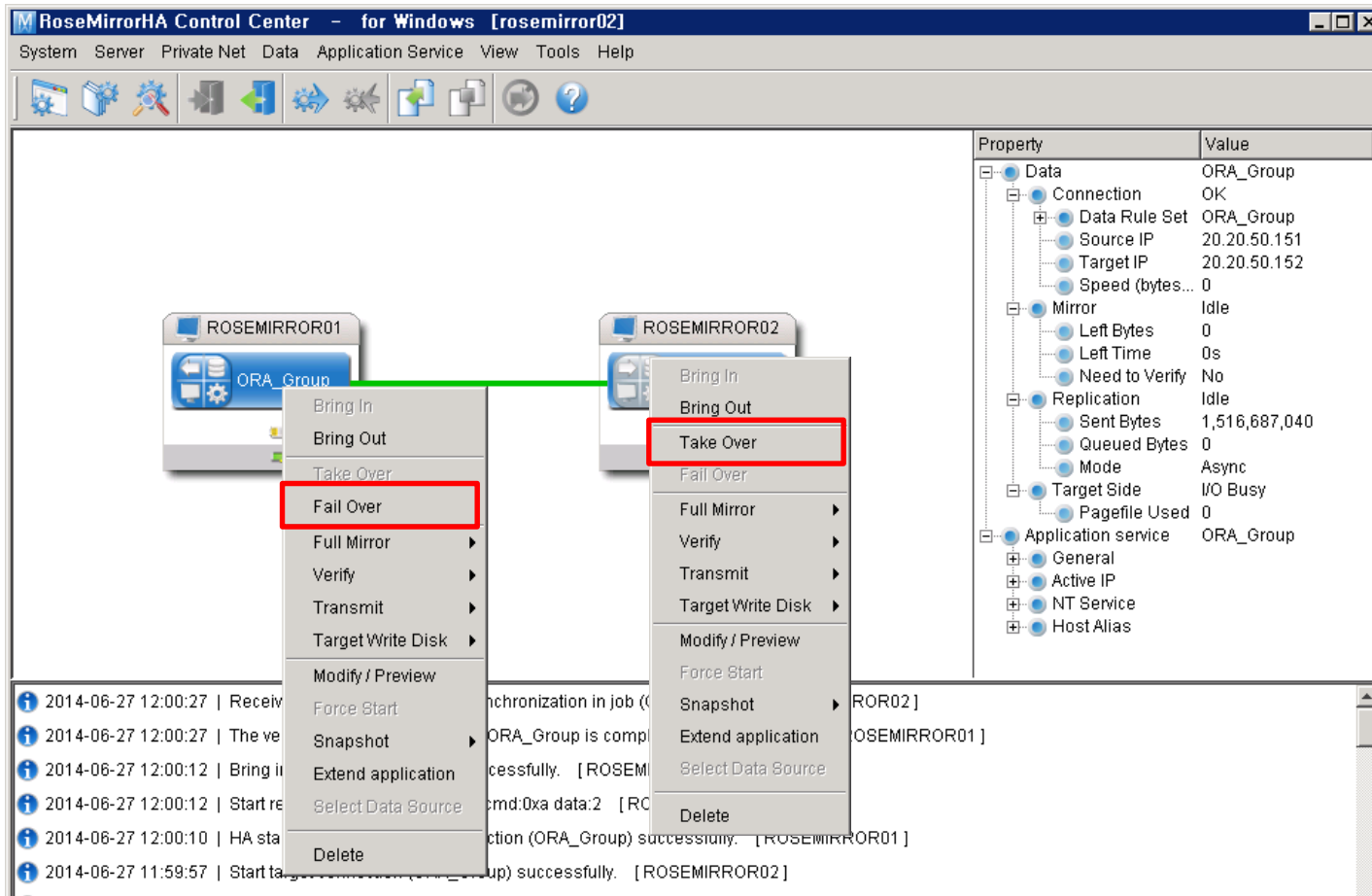
Log messages:

- 2014-06-27 12:00:27 | Receive the signal of finishing synchronization in job (ORA_Group). [ROSEMIRROR02]
- 2014-06-27 12:00:27 | The verification of the connection ORA_Group is completed, time used: 16s. [ROSEMIRROR01]
- 2014-06-27 12:00:12 | Bring in the job (ORA_Group) successfully. [ROSEMIRROR01]
- 2014-06-27 12:00:12 | Start resource <ORA_Group> ok.cmd:0xa data:2 [ROSEMIRROR01]
- 2014-06-27 12:00:10 | HA starts verification in the connection (ORA_Group) successfully. [ROSEMIRROR01]
- 2014-06-27 11:59:57 | Start target connection (ORA_Group) successfully. [ROSEMIRROR02]
- 2014-06-27 11:59:57 | Create snapshot for the connection (ORA_Group) successfully, comment: (Vol:D Type:Auto). [ROSEMIRROR02]
- 2014-06-27 11:59:55 | HA starts the connection (ORA_Group) successfully. [ROSEMIRROR01]
- 2014-06-27 11:59:55 | Start source connection (ORA_Group) successfully. [ROSEMIRROR01]

4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : Fail Over or Take Over

- Online된 Group을 Active >> Standby node로 소유권 전환 및 수동 절체 명령
- Group에서 오른쪽 마우스 클릭 후 "Fail Over" or "Take Over" 선택



4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : Fail Over or Take Over

- Group0| Active node >> Standby node로 수동절체된 화면

The screenshot displays the RoseMirrorHA Control Center window for Windows, titled "RoseMirrorHA Control Center - for Windows [rosemirror02]". The interface includes a menu bar (System, Server, Private Net, Data, Application Service, View, Tools, Help) and a toolbar with various icons. The main workspace shows two server nodes, ROSEMIRROR01 and ROSEMIRROR02, each with an "ORA_Group" resource. A green line connects the "ORA_Group" of ROSEMIRROR01 to the "ORA_Group" of ROSEMIRROR02, indicating a replication or failover path. The right-hand pane shows the "Property" list for the selected "ORA_Group" on ROSEMIRROR02, with values such as "Data: ORA_Group", "Connection: OK", "Data Rule Set: ORA_Group", "Source IP: 20.20.50.152", "Target IP: 20.20.50.151", "Speed (bytes...): 33,064", "Mirror: Idle", "Left Bytes: 0", "Left Time: 0s", "Need to Verify: No", "Replication: Replicating", "Sent Bytes: 2,187,200", "Queued Bytes: 0", "Mode: Async", "Target Side: Idle", "Pagefile Used: 0", and "Application service: ORA_Group". The bottom pane shows a log of events, including a successful failover operation on 2014-06-27 12:18:31.

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.152
Target IP	20.20.50.151
Speed (bytes...)	33,064
Mirror	Idle
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Replicating
Sent Bytes	2,187,200
Queued Bytes	0
Mode	Async
Target Side	Idle
Pagefile Used	0
Application service	ORA_Group
General	
Active IP	
NT Service	
Host Alias	

2014-06-27 12:18:31 | Take over the job (ORA_Group) successfully. [ROSEMIRROR02]
2014-06-27 12:18:31 | Start resource <ORA_Group> ok.cmd:0xc data:1 [ROSEMIRROR02]
2014-06-27 12:18:04 | Start target connection (ORA_Group) successfully. [ROSEMIRROR01]
2014-06-27 12:18:04 | Create snapshot for the connection (ORA_Group) successfully, comment: (Vol:D Type:Auto). [ROSEMIRROR01]
2014-06-27 12:18:02 | HA starts replicating in the connection (ORA_Group) successfully. [ROSEMIRROR02]
2014-06-27 12:18:02 | HA starts the connection (ORA_Group) successfully. [ROSEMIRROR02]
2014-06-27 12:18:02 | Start source connection (ORA_Group) successfully. [ROSEMIRROR02]
2014-06-27 12:17:49 | Add IP <192.168.5.153> to NIC<(316AA514-30CC-4D92-8455-4955D49B21F5)> successfully. [ROSEMIRROR02]
2014-06-27 12:17:41 | Fail over the job (ORA_Group) successfully. [ROSEMIRROR01]

4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : Bring Out

- Group에 등록된 Resources를 Offline
- Group에서 오른쪽 마우스 클릭 후 "Bring Out"선택
- Active - Standby node 중 Group이 Online된 상태에서 가능

The screenshot displays the RoseMirrorHA Control Center for Windows. The main window shows two nodes, ROSEMIRROR01 and ROSEMIRROR02, connected by a green line. The ROSEMIRROR01 node has a context menu open over it, with the 'Bring Out' option highlighted in a red box. The menu also includes options like 'Bring In', 'Take Over', 'Fail Over', 'Full Mirror', 'Verify', 'Transmit', 'TargetWrite Disk', 'Modify / Preview', 'Force Start', 'Snapshot', 'Extend application', 'Select Data Source', and 'Delete'.

On the right side, the 'Property' pane shows the configuration for the selected 'ORA_Group' resource. The properties are as follows:

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.152
Target IP	20.20.50.151
Speed (bytes/s)	33,064
Mirror	Idle
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Idle
Sent Bytes	4,636,696
Queued Bytes	0
Mode	Async
Target Side	I/O Busy
Pagefile Used	0
Application service	ORA_Group
General	
Active IP	
NT Service	
Host Alias	

The bottom pane shows a log of events:

- 2014-06-27 12:18:31 | Take over the job (ORA_Group) successfully. [ROSEMIRROR01]
- 2014-06-27 12:18:31 | Start resource <ORA_Group> ok.cmd:0xc data:1 [ROSEMIRROR01]
- 2014-06-27 12:18:04 | Start target connection (ORA_Group) successfully. [ROSEMIRROR01]
- 2014-06-27 12:18:04 | Create snapshot for the connection (ORA_Group) successfully, c [ROSEMIRROR01]
- 2014-06-27 12:18:02 | HA starts replicating in the connection (ORA_Group) successfully [ROSEMIRROR01]
- 2014-06-27 12:18:02 | HA starts the connection (ORA_Group) successfully. [ROSEMIRROR01]

4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : Bring Out

- Group이 정상 Offline된 후의 화면

The screenshot displays the RoseMirrorHA Control Center for Windows. The main workspace shows two nodes, ROSEMIRROR01 and ROSEMIRROR02, each with an 'ORA_Group' resource. They are connected by a line, indicating a replication or mirroring relationship. The right-hand pane shows the properties of the selected 'ORA_Group' resource, including Data, Connection, Data Rule Set, Mirror, Replication, and Application service settings. The bottom log window shows a series of status messages indicating successful operations and resource management.

Property	Value
Data	ORA_Group
Connection	Stop
Data Rule Set	ORA_Group
Source IP	20.20.50.152
Target IP	20.20.50.151
Speed (bytes...)	0
Mirror	Stop
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Stop
Sent Bytes	6,942,552
Queued Bytes	0
Mode	Async
Target Side	Stop
Pagefile Used	0
Application service	ORA_Group
General	
Active IP	
NT Service	
Host Alias	

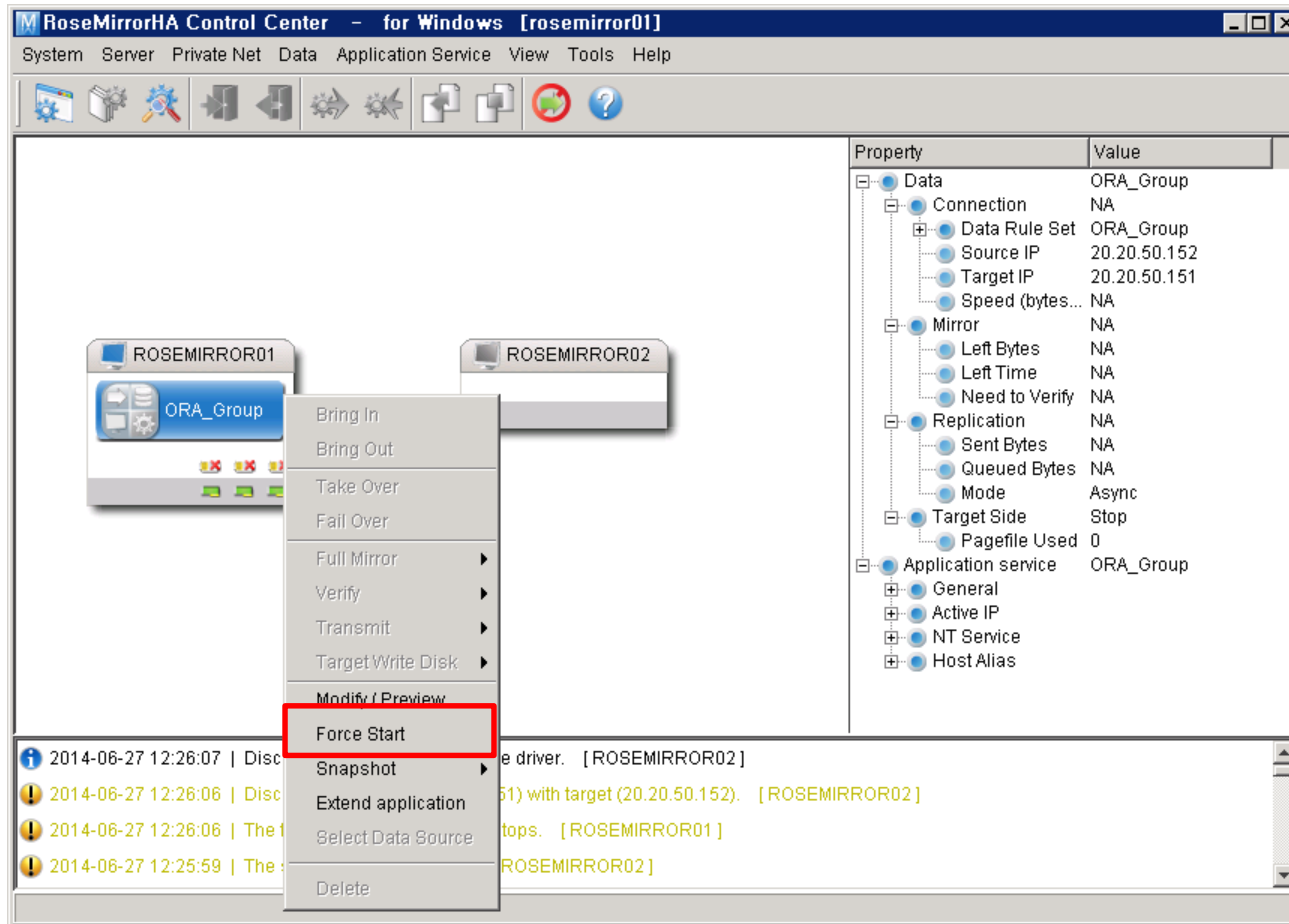
Log messages:

- 2014-06-27 12:24:04 | Bring out the job (ORA_Group) successfully. [ROSEMIRROR02]
- 2014-06-27 12:24:04 | Bring out the job (ORA_Group) successfully. [ROSEMIRROR01]
- 2014-06-27 12:24:04 | Stop resource <ORA_Group> ok.cmd:0xb data:1 [ROSEMIRROR01]
- 2014-06-27 12:24:04 | Stop resource <ORA_Group> begin.<cmd:0xb reason:8 state:1 data:1> [ROSEMIRROR01]
- 2014-06-27 12:24:03 | Stop resource <ORA_Group> ok.cmd:0xb data:1 [ROSEMIRROR02]
- 2014-06-27 12:24:03 | Del IP <192.168.5.153> from NIC-<{316AA514-30CC-4D92-8455-4955D49B21F5}> successfully. [ROSEMIRROR02]
- 2014-06-27 12:24:02 | When the remote notification job (ORA_Group) stops, data are ok. [ROSEMIRROR01]
- 2014-06-27 12:24:01 | Stop Job (ORA_Group) and connection data are ok. [ROSEMIRROR02]
- 2014-06-27 12:24:00 | The target connection (ORA_Group) is stopped successfully. [ROSEMIRROR01]

4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : Force Start

- Group을 정상으로 Bring In할 수 없는 경우에 가능(정상 상태에서는 비활성화)
- Group에서 오른쪽 마우스 클릭 후 "Force Start" 선택
- Active - Standby node 중 하나만 운영가능한 상태에서 가능



4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : Force Start

- Group이 Force Start를 진행한 후의 화면

The screenshot displays the RoseMirrorHA Control Center for Windows. The main window shows two server instances: ROSEMIRROR01 and ROSEMIRROR02. The ROSEMIRROR01 instance is highlighted with a red box, showing the 'ORA_Group' resource. The right pane displays the properties of the selected resource, and the bottom pane shows the event log.

Property	Value
Data	ORA_Group
Connection	Stop
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	0
Mirror	Stop
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Stop
Sent Bytes	1,528,960,184
Queued Bytes	0
Mode	Async
Target Side	Absent
Pagefile Used	NA
Application service	ORA_Group
General	
Active IP	
NT Service	
Host Alias	

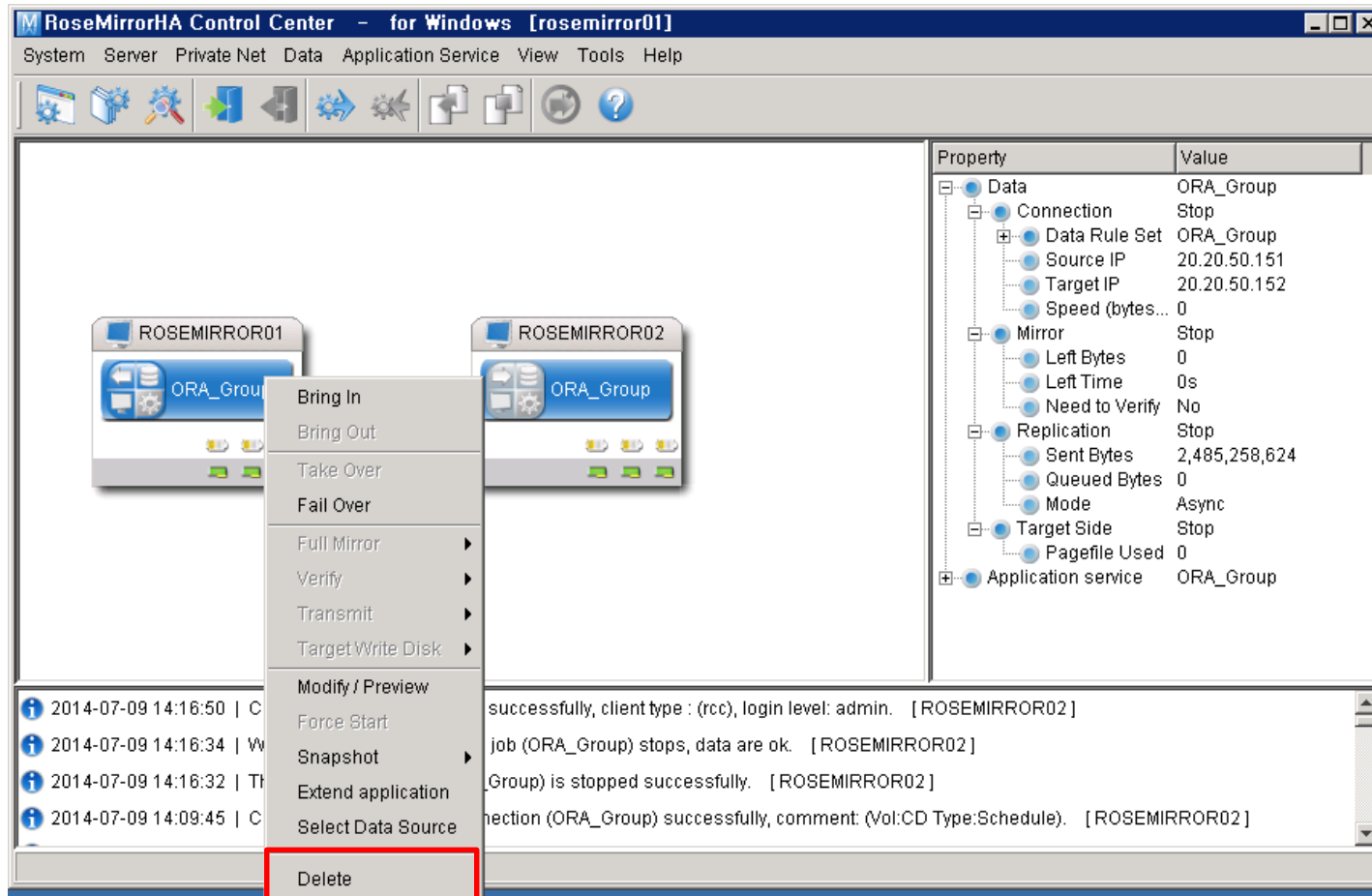
Event Log:

- 2014-06-27 14:01:23 | Start resource <ORA_Group> ok.cmd:0x21 data:4 [ROSEMIRROR01]
- 2014-06-27 14:01:16 | The server finishes dealing with shutdown event. [ROSEMIRROR01]
- 2014-06-27 14:01:16 | Not wait volume during shutdown takeover, because remote shutdown is normal. [ROSEMIRROR01]
- 2014-06-27 14:01:16 | The server begins to deal with shutdown event. [ROSEMIRROR01]

4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : Delete

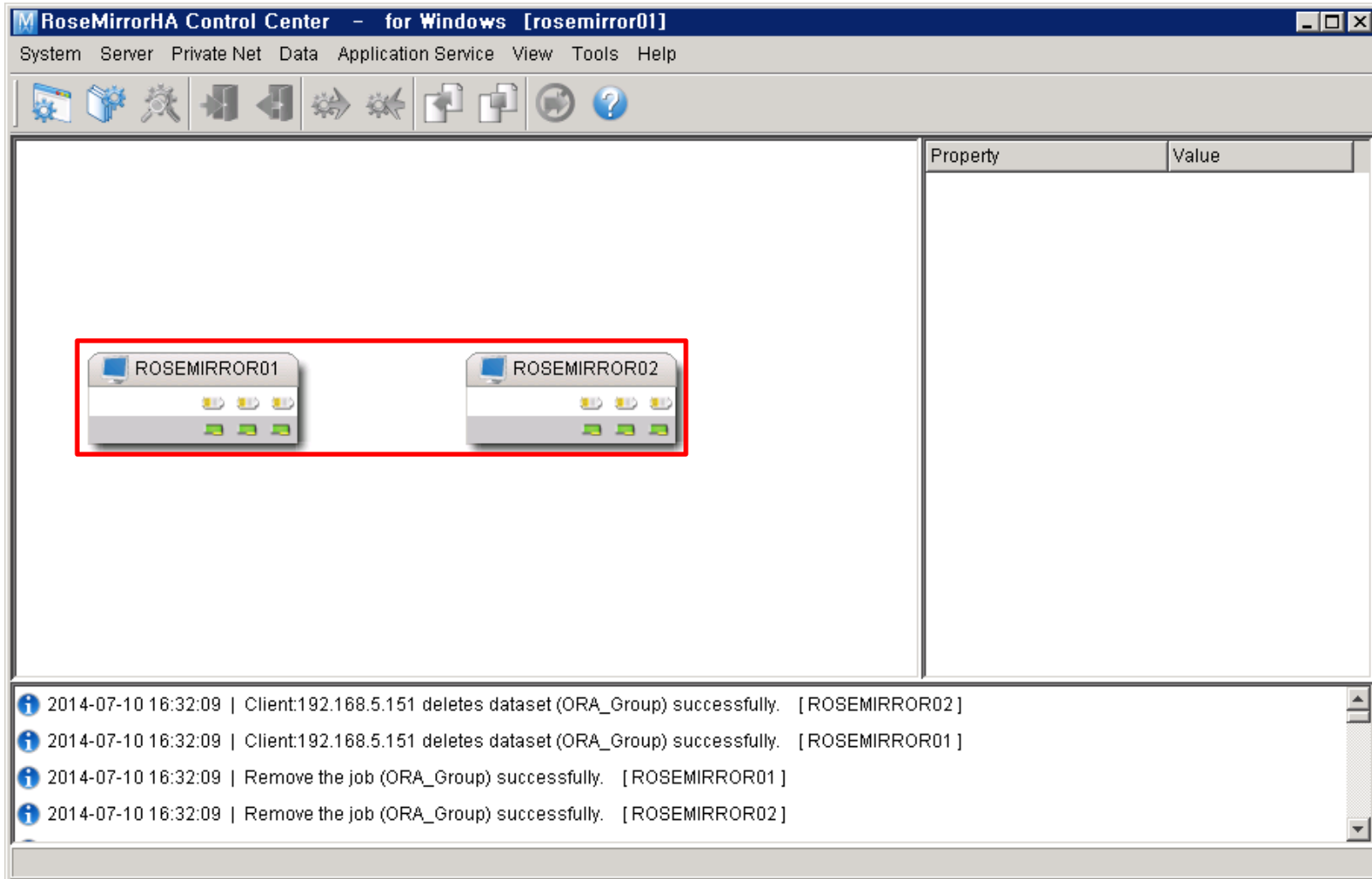
- Group을 삭제함
- 운영서비스를 중지해야 가능함



4. RoseMirroHA 그룹 조작

◆ RoseMirrorHA : Delete

- Group을 삭제한 화면



5. Replication Operation

- Replication : Full Mirror
- Replication : verify
- Replication : Transmit
- Replication : TargetWriteDisk
- Replication : Select Source

5. Replication Operation

◆ Replication Operation : Full Mirror

- Replication Data 전체를 처음부터 다시 진행
- Group에서 오른쪽 마우스 클릭 후 "Full Mirror" 선택
- Start, Stop, Pause, Resume 가능

The screenshot displays the RoseMirrorHA Control Center interface. In the center, two server icons, ROSEMIRROR01 and ROSEMIRROR02, are connected by a green line. ROSEMIRROR01 is labeled 'ORA_Group' and ROSEMIRROR02 is labeled 'ORA_Group'. A right-click context menu is open over ROSEMIRROR01, with 'Full Mirror' selected. A sub-menu is visible for 'Full Mirror', containing 'Start', 'Stop', 'Pause', and 'Resume'. The 'Property' pane on the right shows the following details:

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	0
Mirror	Idle
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Idle
Sent Bytes	1,589,745,280
Queued Bytes	0
Mode	Async
Target Side	Idle
Pagefile Used	0
Application service	ORA_Group

The bottom log pane shows the following entries:

- 2014-06-27 15:27:48 | Login
- 2014-06-27 15:27:45 | Login
- 2014-06-27 15:26:52 | Serve
- 2014-06-27 15:26:52 | Serve

5. Replication Operation

◆ Replication Operation : Full Mirror

- Full Mirror를 진행할 경우 복제를 다시 진행하는 화면

The screenshot shows the RoseMirrorHA Control Center interface. On the left, two nodes, ROSEMIRROR01 and ROSEMIRROR02, are connected by a green line. Both nodes show an 'ORA_Group' icon. On the right, a tree view displays the properties of the selected 'Data' group. The 'Mirror' property is highlighted with a red box, showing 'Mirror(43%)'. Other properties include 'Left Bytes' (911,126,528), 'Left Time' (9s), and 'Need to Verify' (No).

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	93,562,496
Mirror	Mirror(43%)
Left Bytes	911,126,528
Left Time	9s
Need to Verify	No
Replication	Idle
Sent Bytes	3,904,120,952
Queued Bytes	2,048
Mode	Async
Target Side	Full Mirror
Pagefile Used	0
Application service	ORA_Group

- Mirror(43%) 과 같이 Full Mirror 진행 상황을 알 수 있다.

5. Replication Operation

◆ Replication Operation : Verify

- Active node의 Data와 Standby node의 Data의 정합성을 확인
- Group에서 오른쪽 마우스 클릭 후 "Verify" 선택
- Start, Stop 가능

The screenshot displays the RoseMirrorHA Control Center interface. On the left, two nodes are shown: ROSEMIRROR01 and ROSEMIRROR02. A context menu is open over the ORA_Group on ROSEMIRROR02, with the 'Verify' option selected. A sub-menu is visible with 'Start' and 'Stop' options. On the right, the 'Property' pane shows the status of the ORA_Group. The 'Data' property is expanded, showing 'Connection' as OK, 'Data Rule Set' as ORA_Group, 'Source IP' as 20.20.50.151, 'Target IP' as 20.20.50.152, and 'Speed (bytes...)' as 0. The 'Mirror' property is expanded, showing 'Left Bytes' as 0, 'Left Time' as 0s, and 'Need to Verify' as No. The 'Replication' property is expanded, showing 'Sent Bytes' as 4,820,196,552, 'Queued Bytes' as 0, and 'Mode' as Async. The 'Target Side' property is expanded, showing 'Pagefile Used' as 0. The 'Application service' property is expanded, showing 'ORA_Group'.

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...)	0
Mirror	Idle
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Idle
Sent Bytes	4,820,196,552
Queued Bytes	0
Mode	Async
Target Side	Idle
Pagefile Used	0
Application service	ORA_Group

Log messages at the bottom:

- 2014-06-27 15:42:12 | Recei...
- 2014-06-27 15:42:10 | The m...
- 2014-06-27 15:41:52 | Client...
- 2014-06-27 15:39:54 | Recei...
- 2014-06-27 15:39:54 | The m...

5. Replication Operation

◆ Replication Operation : Verify

- Verify를 진행할 경우 정합성을 다시 확인하는 화면

The screenshot displays the RoseMirrorHA Control Center interface. On the left, two nodes, ROSEMIRROR01 and ROSEMIRROR02, are shown with a green line connecting them, indicating a replication link. Both nodes have an 'ORA_Group' icon. On the right, a tree view shows the properties of the replication setup. The 'Mirror' property is highlighted with a red box, showing the following details:

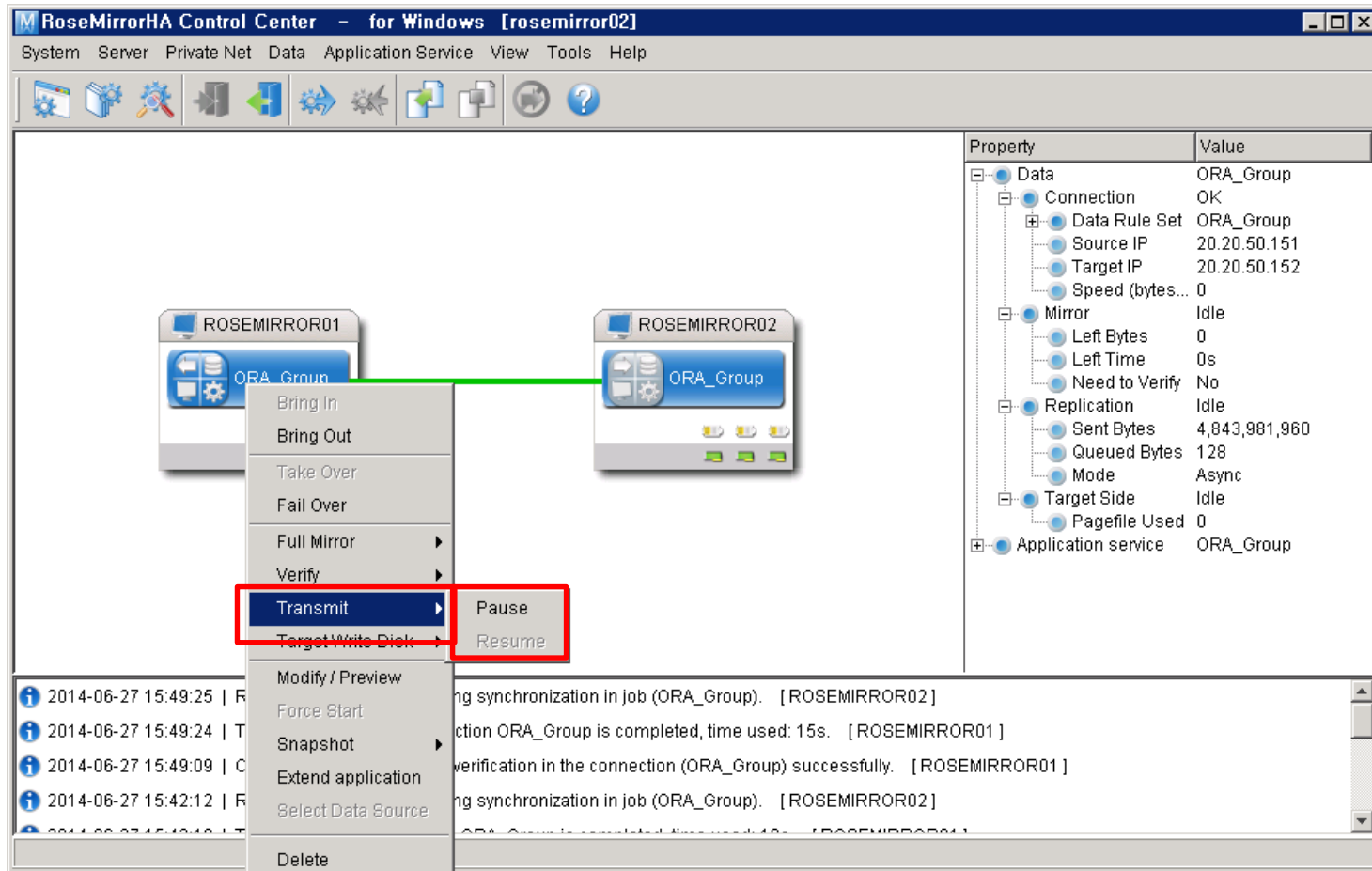
Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	93,562,496
Mirror	Mirror(43%)
Left Bytes	911,126,528
Left Time	9s
Need to Verify	No
Replication	Idle
Sent Bytes	3,904,120,952
Queued Bytes	2,048
Mode	Async
Target Side	Full Mirror
Pagefile Used	0
Application service	ORA_Group

- Verifying(45%)과 같이 진행 상황을 알 수 있다.

5. Replication Operation

◆ Replication Operation : Transmit

- Replication Area(Volume, Folder, File)에 추가, 수정, 삭제된 Data를 Standby node로 전달하는 기능
- Group에서 오른쪽 마우스 클릭 후 "Transmit" 선택
- Pause Resume 가능



5. Replication Operation

◆ Replication Operation : Transmit

- Transmit를 진행할 경우의 화면

The screenshot displays the RoseMirrorHA Control Center interface. The main window shows two mirror nodes, ROSEMIRROR01 and ROSEMIRROR02, connected by a yellow line. The 'ORA_Group' is selected on both nodes. The right pane shows the 'Property' list for the selected group. The 'Mirror' section is expanded, and the 'Speed (bytes/sec)' property is highlighted in yellow, indicating a pause. The 'Speed (bytes/sec)' property is set to 'Pause'.

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes/sec)	Pause
Mirror	Idle
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Idle
Sent Bytes	4,844,708,488
Queued Bytes	5,910,672
Mode	Async
Target Side	Idle
Pagefile Used	0
Application service	ORA_Group

- 복제 speed의 일시 중지와 복제 상태의 변경을 확인할 수 있다.(정상 : 녹색, 일시중지 : 노란색)

5. Replication Operation

◆ Replication Operation : TargetWrite Disk

- Standby node의 Replication Area에 “Write”를 일시 중지한다.
- Group에서 오른쪽 마우스 클릭 후 “Target Write Disk” 선택
- Pause, Resume 가능

The screenshot displays the RoseMirrorHA Control Center interface. On the left, two nodes, ROSEMIRROR01 and ROSEMIRROR02, are shown with a context menu open over ROSEMIRROR01. The menu includes options like 'Bring In', 'Bring Out', 'Take Over', 'Fail Over', 'Full Mirror', 'Verify', 'Transmit', 'Target Write Disk', 'Modify / Preview', 'Force Start', 'Snapshot', 'Extend application', 'Select Data Source', and 'Delete'. The 'Target Write Disk' option is highlighted, and a sub-menu is visible with 'Pause' and 'Resume' options. On the right, the 'Property' pane shows details for the 'ORA_Group' replication, including connection status (OK), source and target IP addresses, speed, and replication mode (Async). The bottom status bar shows a log of recent events.

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...)	33,064
Mirror	Idle
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Replicating
Sent Bytes	4,859,020,000
Queued Bytes	0
Mode	Async
Target Side	Idle
Pagefile Used	0
Application service	ORA_Group

5. Replication Operation

- ◆ Replication Operation : TargetWrite Disk
 - Target Write Disk를 진행할 경우의 화면

The screenshot displays the RoseMirrorHA Control Center interface. On the left, two server icons, ROSEMIRROR01 and ROSEMIRROR02, are connected by a green line. ROSEMIRROR02 has a red box around its 'ORA_Group' icon. On the right, a 'Property' table shows the configuration for the selected 'ORA_Group'.

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...)	0
Mirror	Idle
Left Bytes	0
Left Time	0s
Need to Verify	No
Replication	Idle
Sent Bytes	4,862,484,400
Queued Bytes	0
Mode	Async
Target Side	Pause
Pagefile Used	1,846,100
Application service	ORA_Group

- Target Side의 Pause와  >>  변화를 알 수 있다.

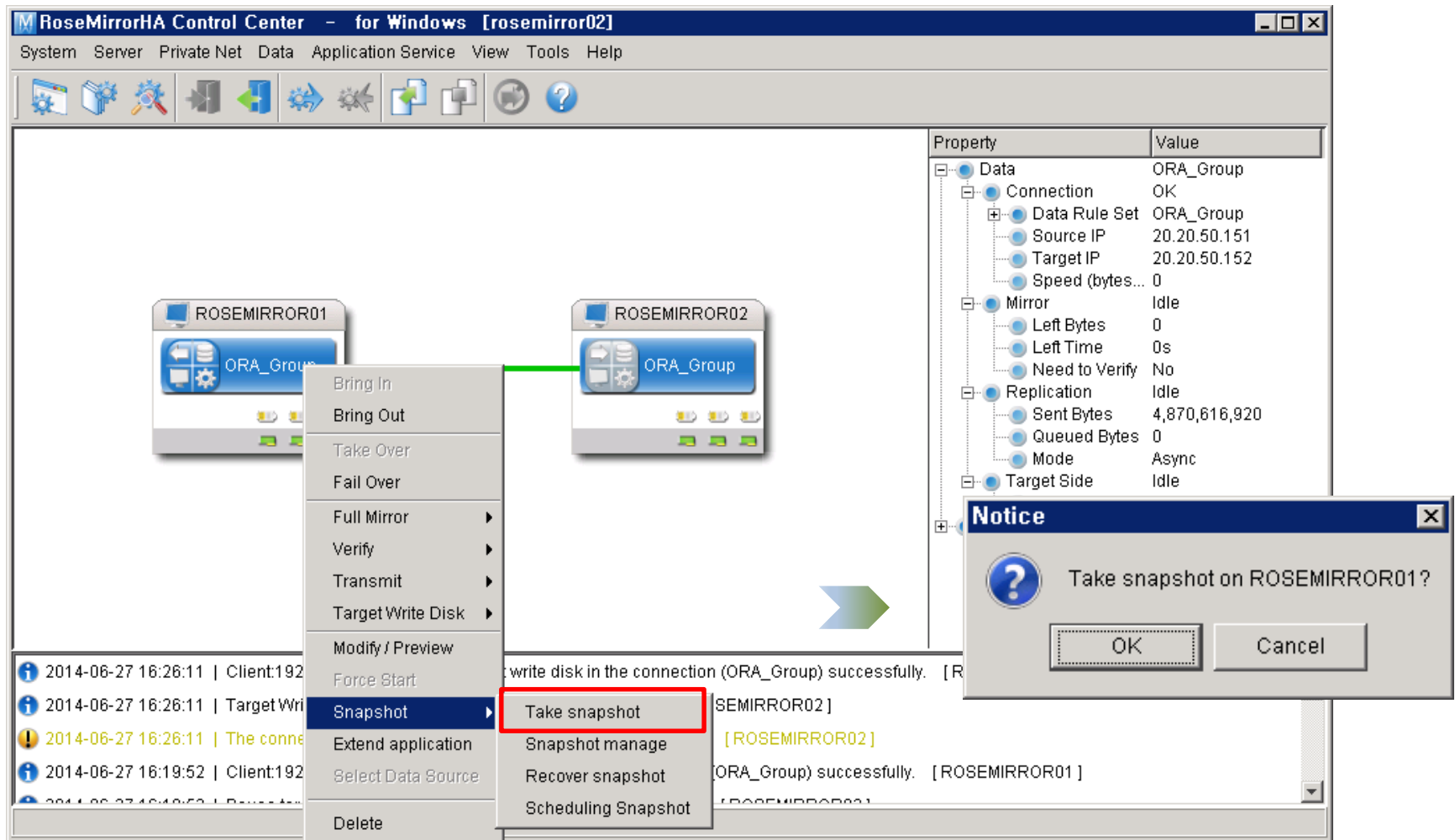
6. Snapshot Operation

- Take Snapshot
- Scheduling Snapshot
- Snapshot Manage
- Recover Snapshot

6. Snapshot Operation

◆ Snapshot Operation : Take Snapshot

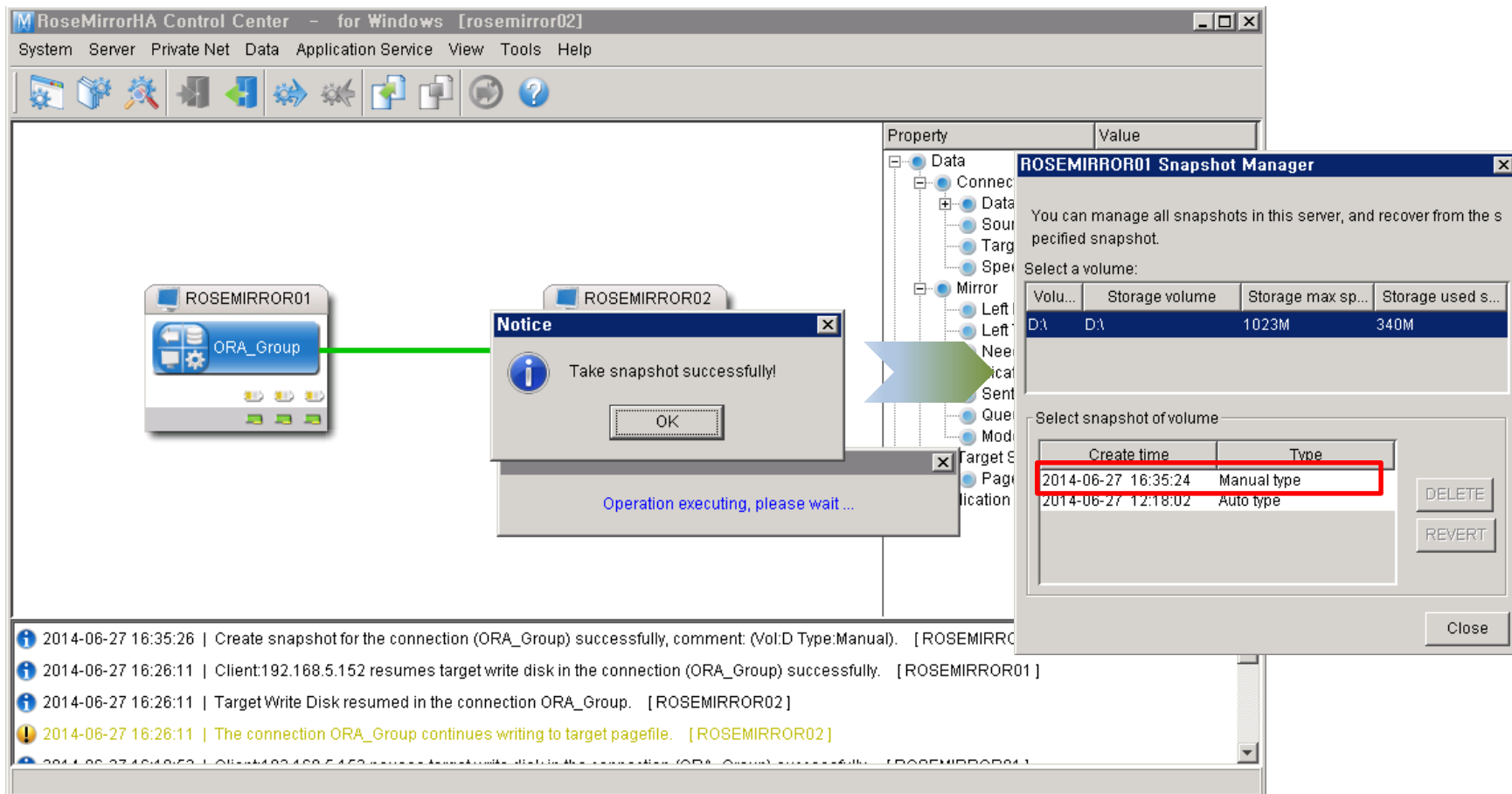
- 운영 중인 Replication Area의 Data를 관리자가 수동으로 Snapshot 한다.
- Group에서 오른쪽 마우스 클릭 후 "Snapshot >> Take Snapshot" 선택



6. Snapshot Operation

◆ Snapshot Operation : Take Snapshot

- Take Snapshot을 진행한 후의 화면



- Snapshot을 성공적으로 진행했다는 팝업을 확인 할 수 있다.

6. Snapshot Operation

◆ Snapshot Operation : Scheduling Snapshot

- Snapshot이 실행될 Schedule을 설정
- Group에서 오른쪽 마우스 클릭 후 "Snapshot >> Scheduling Snapshot" 선택

The screenshot displays the RoseMirrorHA Control Center interface. On the left, two nodes are shown: ROSEMIRROR01 and ROSEMIRROR02, both containing an 'ORA_Group'. A right-click context menu is open over the ROSEMIRROR02 node, with the 'Snapshot' option selected, leading to a sub-menu where 'Scheduling Snapshot' is highlighted with a red box. On the right, the 'Property' pane shows details for the selected 'ORA_Group', including Connection (OK), Data Rule Set (ORA_Group), Source IP (20.20.50.151), Target IP (20.20.50.152), Speed (474,992 bytes/sec), Mirror (Idle), and Left Bytes (0). A 'Snapshot scheme' dialog box is also open, showing the 'Scheduling snapshot' option selected. The dialog includes a description: 'Do snapshot only at appointed point. if data is the same then do snapshot, either will do snapshot after data be same.' It features input fields for 'Snapshot Interval' (0), 'Day' (0), 'Hour' (10), and 'minute' (0). The 'Start now' radio button is selected. The 'Snapshot storage' section shows a table with columns 'Volu...', 'Storage volume', 'Storage max s...', and 'Storage used s...'. The table contains one entry: 'D:\' with 'Storage volume' of 1023M and 'Storage used s...' of 340M. The dialog has 'Set', 'OK', and 'Cancel' buttons.

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...)	474,992
Mirror	Idle
Left Bytes	0

Volu...	Storage volume	Storage max s...	Storage used s...
D:\	D:\	1023M	340M

6. Snapshot Operation

◆ Snapshot Operation : Scheduling Snapshot

- Scheduling Snapshot을 설정하는 화면

Snapshot scheme

☒ Scheduling snapshot

Do snapshot only at appointed point.if data is the same then do snapshot, either will do snapshot after data be same.

Snapshot Interval 0 Day 0 Hour 10 minute

☒ Start now

☐ Start from 2014-06-27 0:00:00

Snapshot storage

Volu...	Storage volume	Storage max s...	Storage used s...
D:\	D:\	1023M	340M

Set

OK Cancel

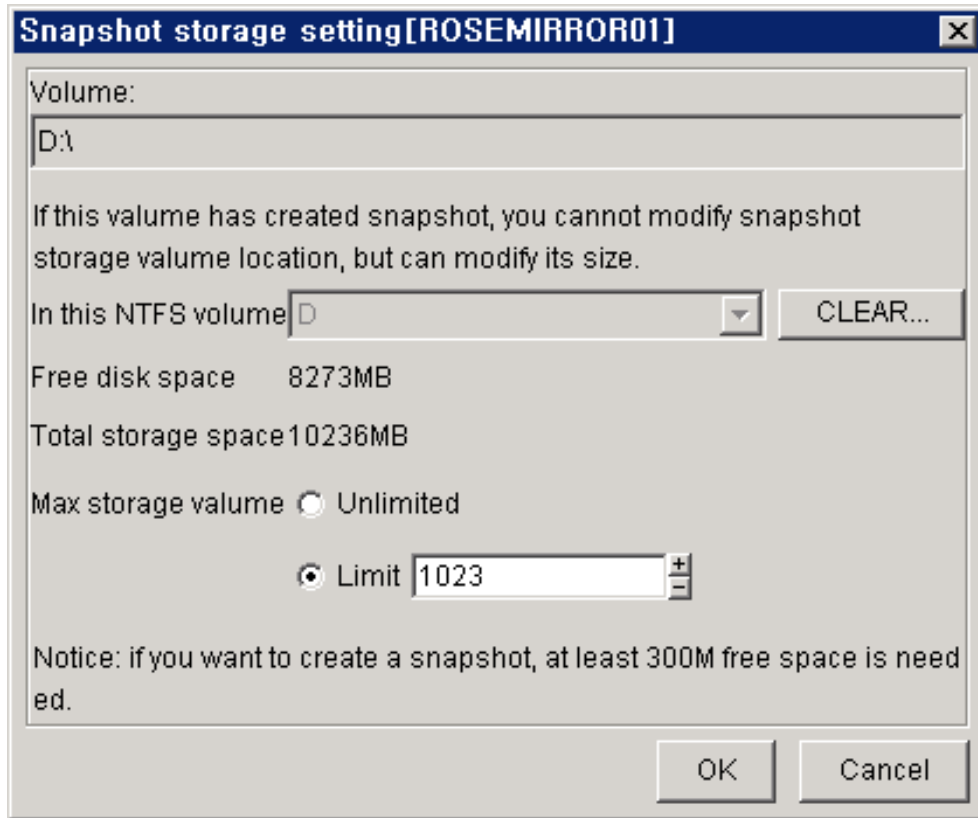
- Snapshot Interval

- Day : 일자
- Hour : 시간
- Minute : 분
- Ex) 화면은 10분간격으로 Snapshot 실행

6. Snapshot Operation

◆ Snapshot Operation : Scheduling Snapshot

- Scheduling Snapshot을 설정하는 화면



• Snapshot Storage Setting

- Snapshot을 저장하는 최대 크기 지정
- Default는 해당 Volume의 10% 사용
- 크기는 변경 가능함

6. Snapshot Operation

◆ Snapshot Operation : Scheduling Manage

- Take Snapshot, Scheduling Snapshot, Auto Snapshot을 관리자가 삭제할 수 있음

The screenshot displays the RoseMirrorHA Control Center interface for Windows. The main window shows two server icons, ROSEMIRROR01 and ROSEMIRROR02, both with 'ORA_Group' data. A context menu is open over ROSEMIRROR01, with the 'Snapshot' option selected, leading to a sub-menu where 'Snapshot manage' is highlighted. A green arrow points from the 'Snapshot manage' option to the 'ROSEMIRROR01 Snapshot Manager' dialog box. The dialog box contains a table of storage volumes and a list of snapshots.

Property Value

Property	Value
Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	474,992
Mirror	Idle
Left Bytes	0
Left Time	0s
N	ROSEMIRROR01 Snapshot Manager
Repli	
S	
Q	
M	
Target	
P	
Applicati	

ROSEMIRROR01 Snapshot Manager

You can manage all snapshots in this server, and recover from the specified snapshot.

Select a volume:

Volu...	Storage volume	Storage max sp...	Storage used s...
D:\	D:\	1023M	921M
C:\	C:\	2447M	381M

Select snapshot of volume

Create time	Type
2014-07-08 15:17:02	Manual type
2014-07-08 12:26:49	Schedule type
2014-07-08 12:26:46	Auto type
2014-07-04 9:50:47	Schedule type
2014-07-04 8:40:44	Schedule type
2014-07-04 8:30:40	Schedule type

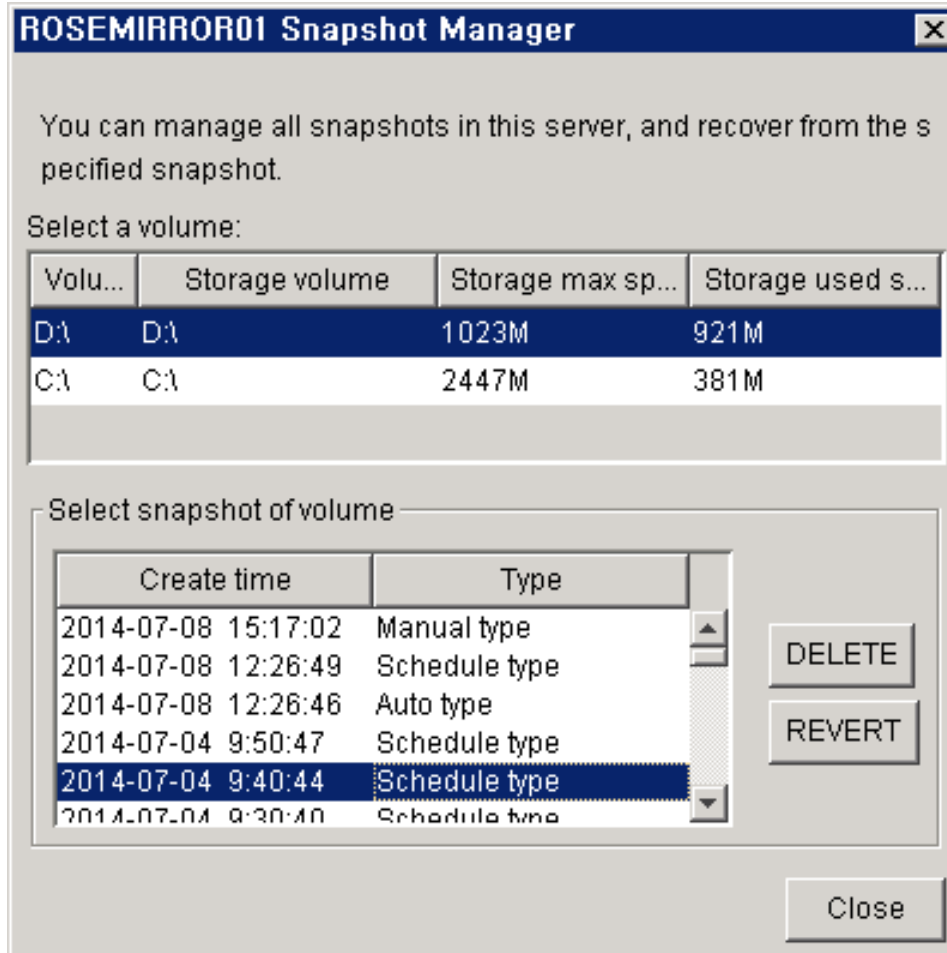
DELETE REVERT

Close

6. Snapshot Operation

◆ Snapshot Operation : Scheduling Manage

- Scheduling Snapshot을 설정하는 화면

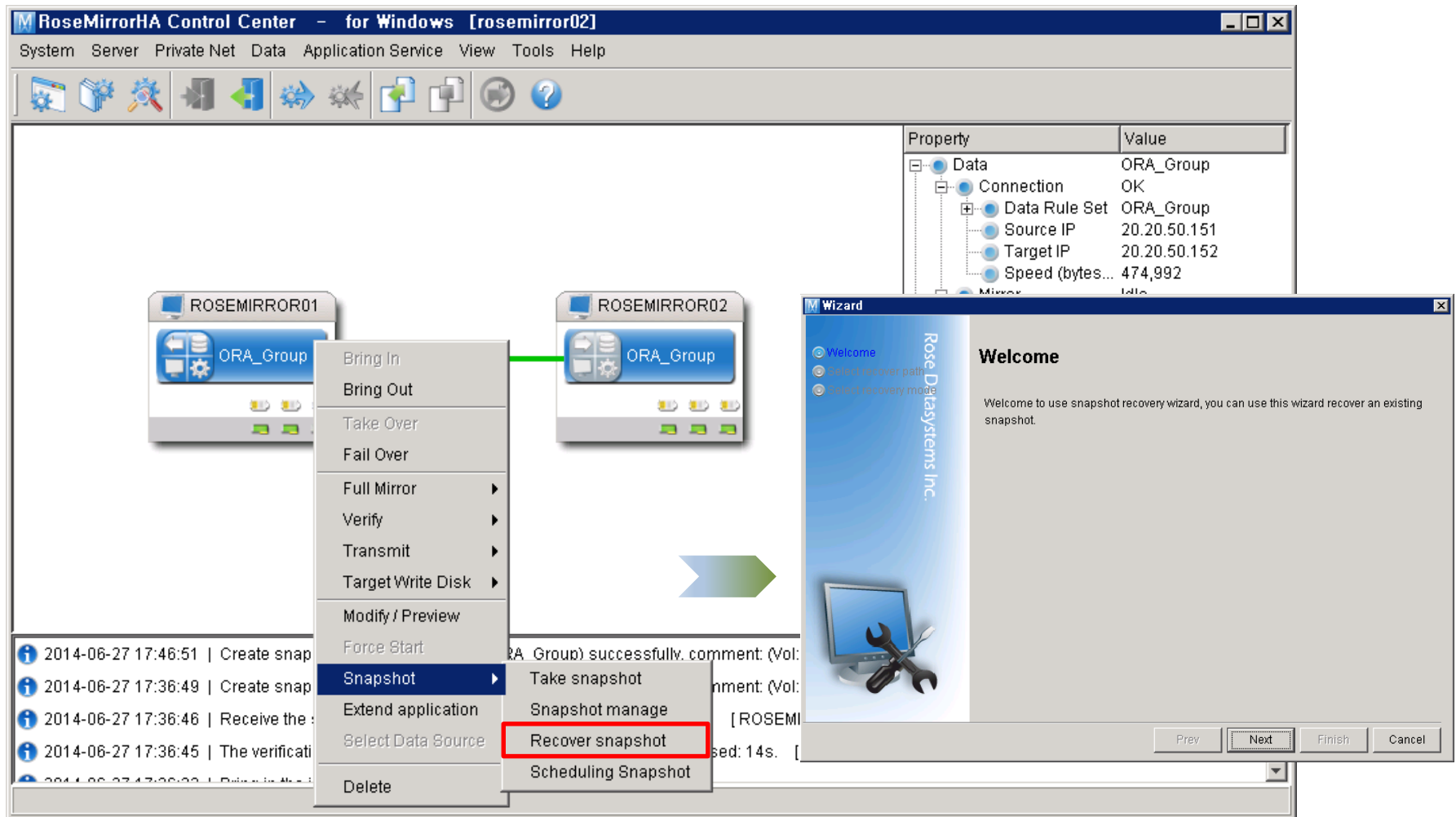


- DELETE
 - 선택한 Snapshot을 삭제함
- REVERT
 - 절대 사용 금지

6. Snapshot Operation

◆ Snapshot Operation : Recover Snapshot(1)

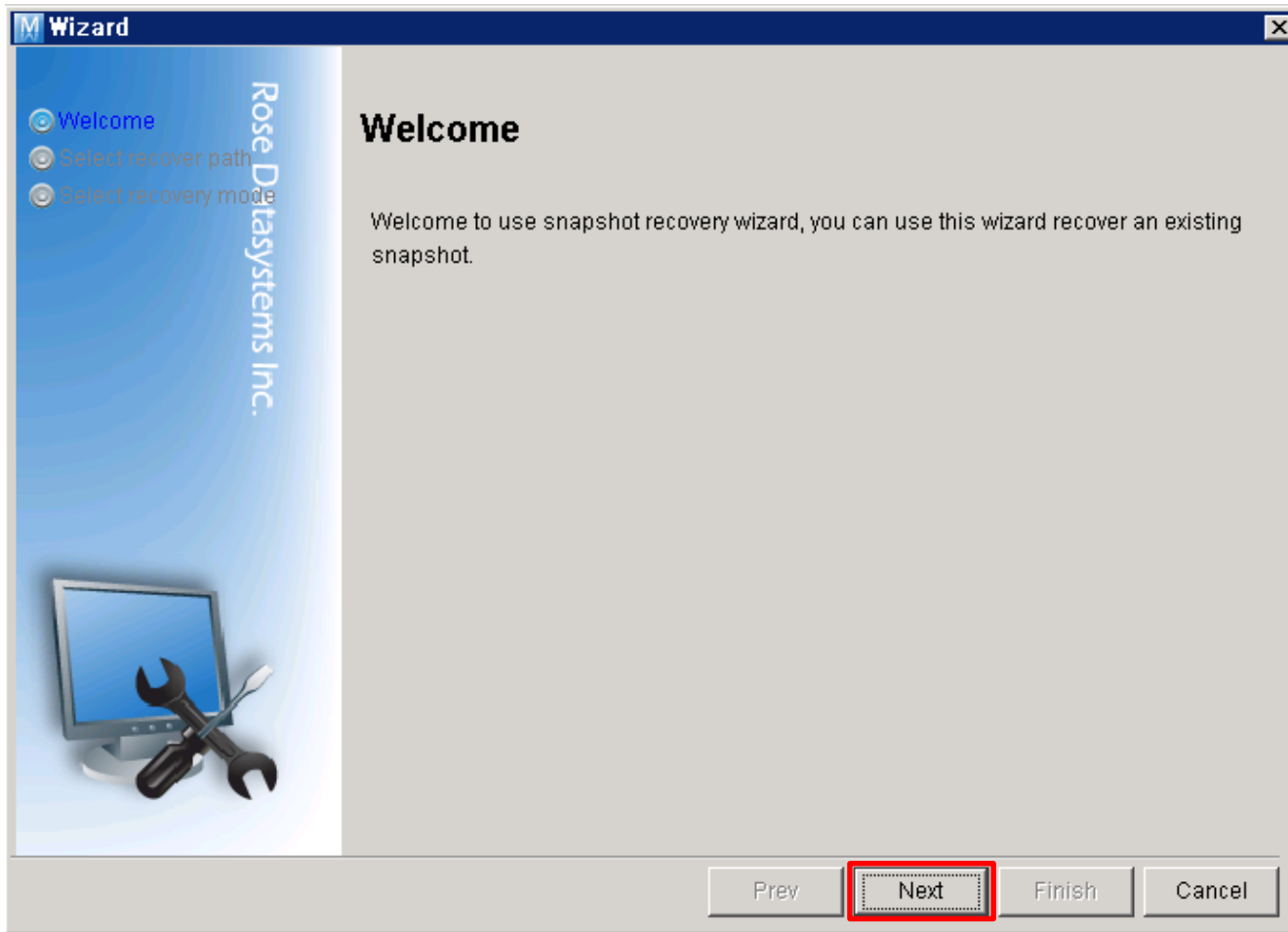
- Take Snapshot, Scheduling Snapshot, Auto Snapshot으로 저장된 Snapshot으로 Replication Data를 복구함
- **주의사항 : 운영서비스 중지가 필요함**



6. Snapshot Operation

◆ Snapshot Operation : Recover Snapshot(2)

- 주의사항 : 운영서비스 중지가 필요함. 즉 Downtime 필요함

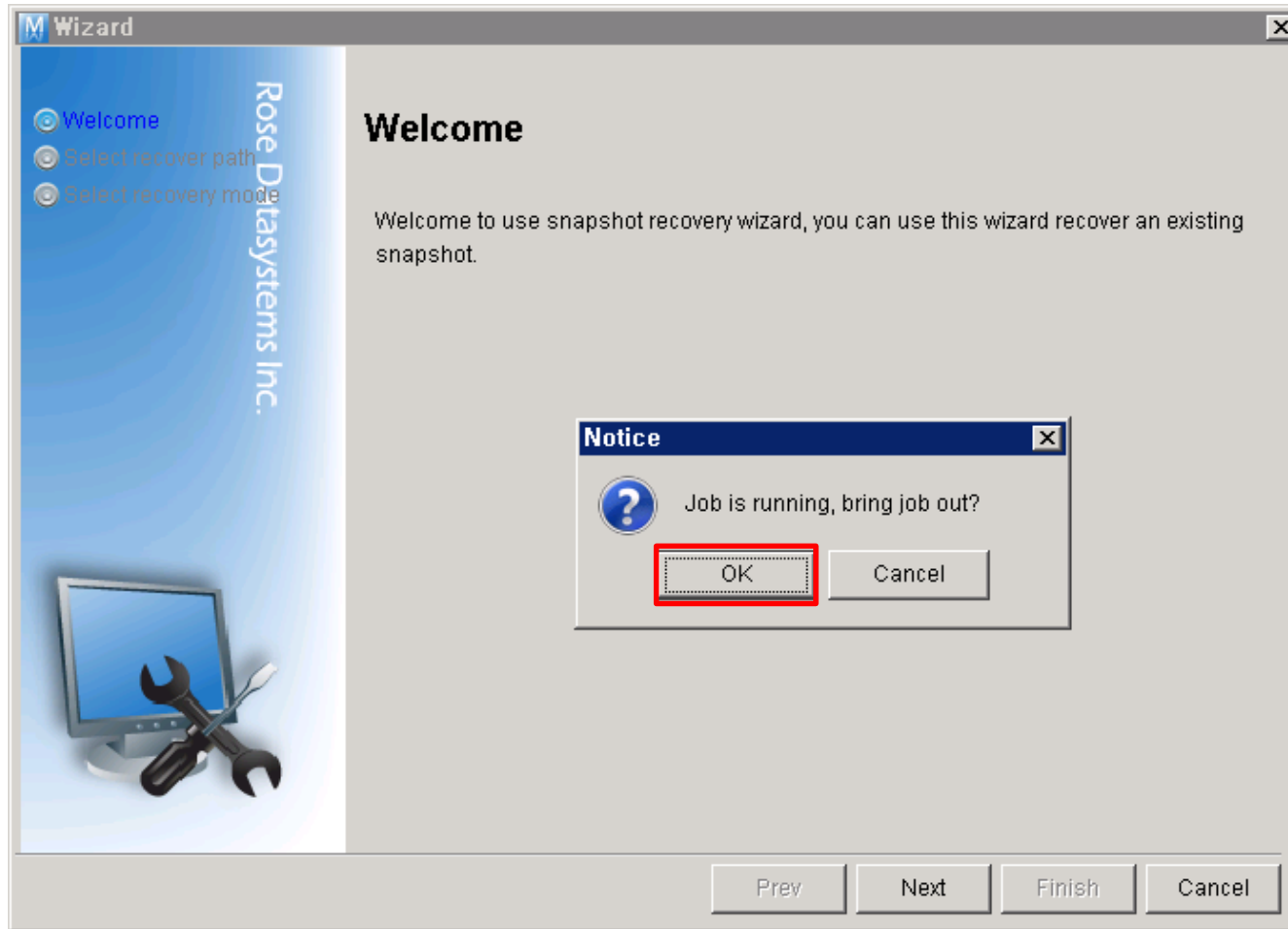


- Next 선택

6. Snapshot Operation

◆ Snapshot Operation : Recover Snapshot(3)

- 주의사항 : 운영서비스 중지가 필요함. 즉 Downtime 필요함

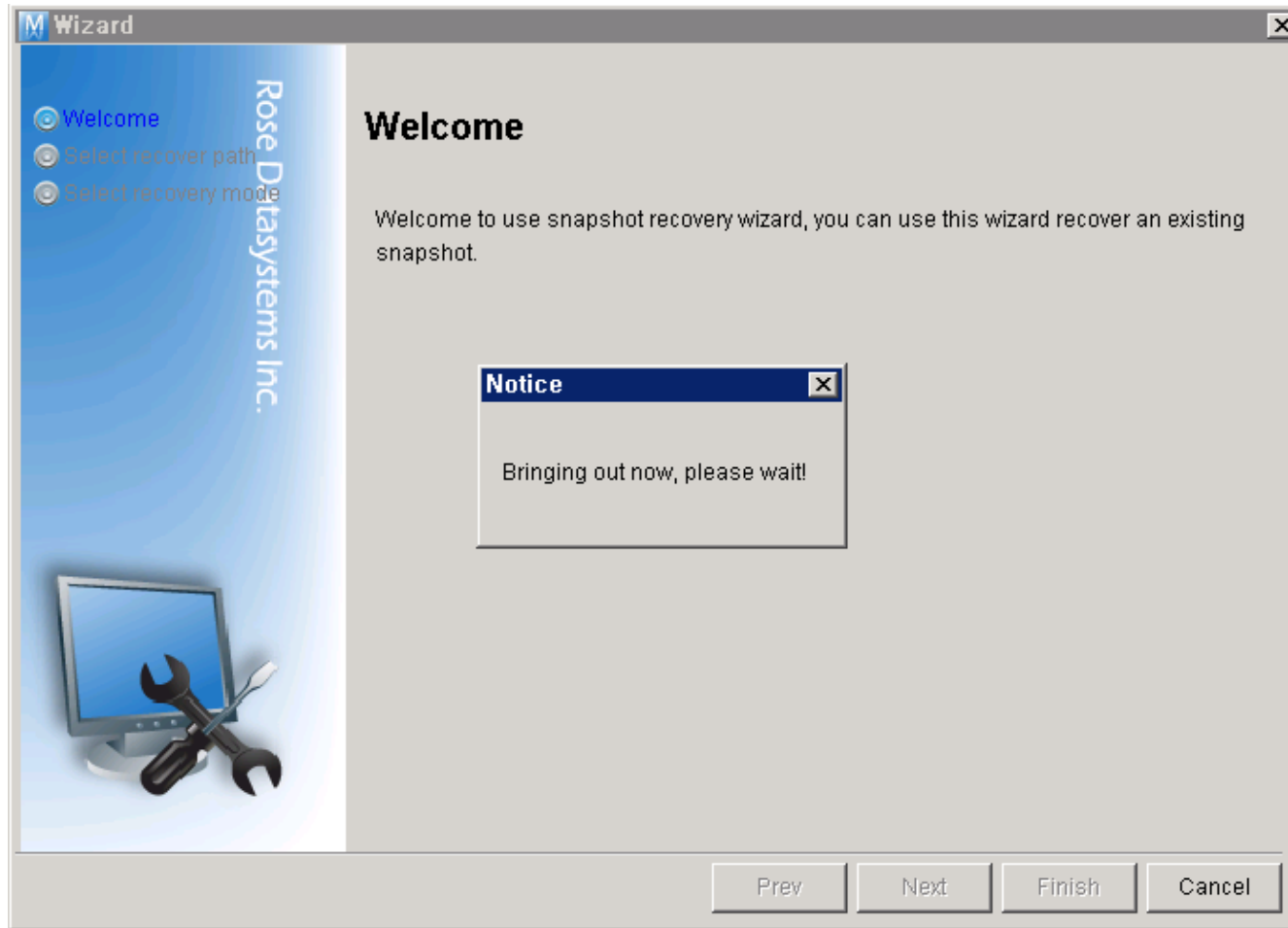


- OK 선택

6. Snapshot Operation

◆ Snapshot Operation : Recover Snapshot(4)

- 주의사항 : 운영서비스 중지가 필요함. 즉 Downtime 필요함

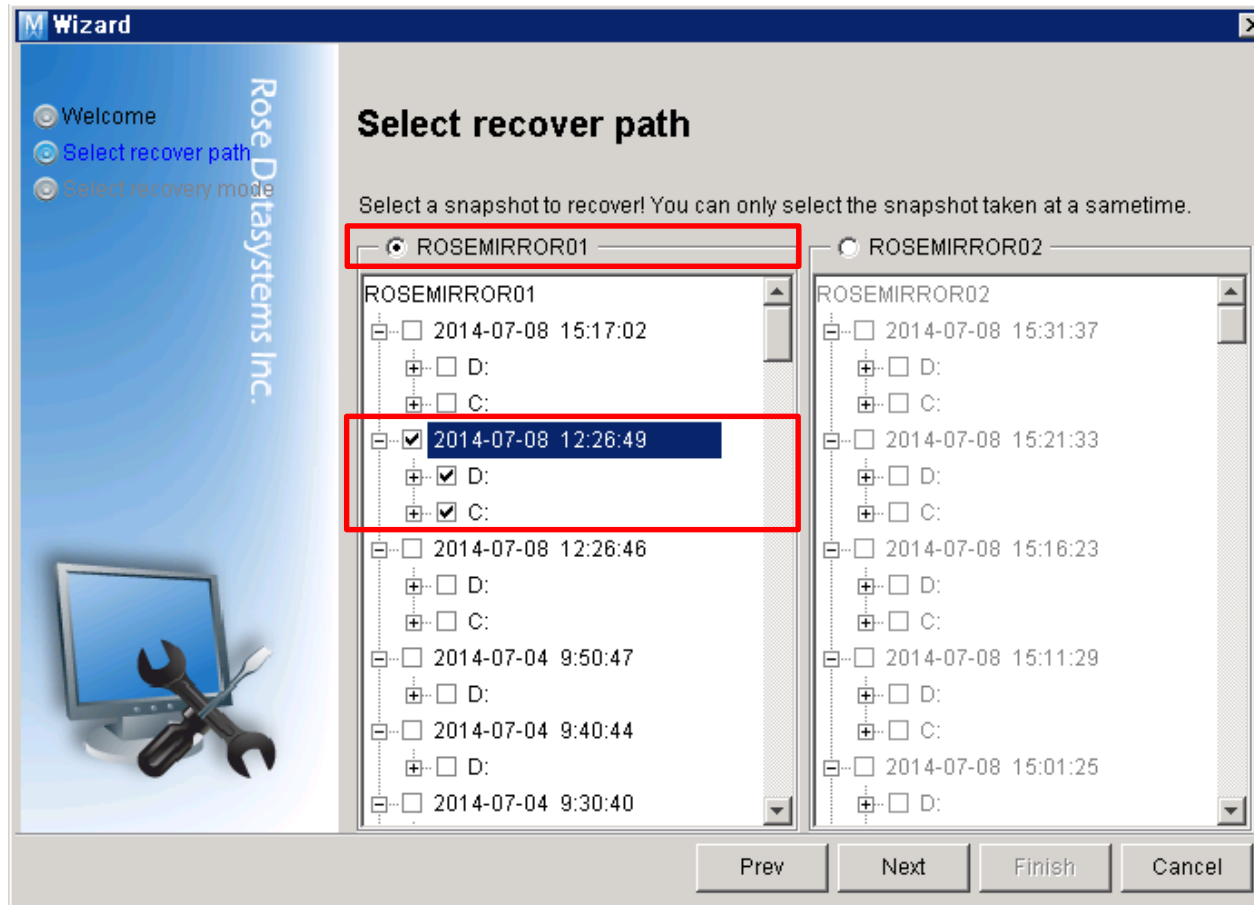


- Group을 Bring Out 진행 중...

6. Snapshot Operation

◆ Snapshot Operation : Recover Snapshot(5)

- 주의사항 : 운영서비스 중지가 필요함. 즉 Downtime 필요함



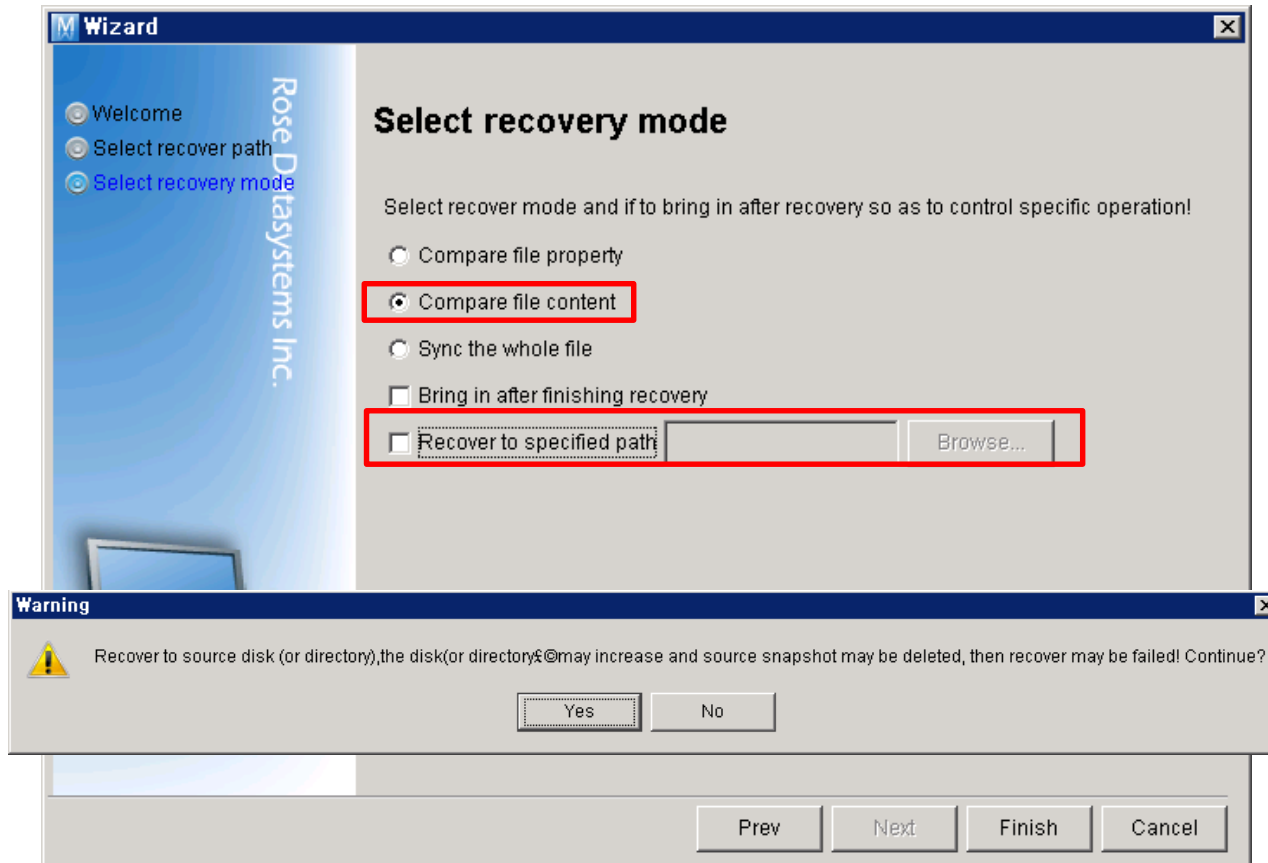
- Node 선택
- 복구시점 선택
- 복구 Data 선택
- Ex) ROSEMIRROR01 nod1
2014.07.08.12:20 경에 모든
Data에서 문제 발생함.
 - 화면과 같이 설정

- Recover를 진행할 node와 시점복구가 필요한 Data를 선택

6. Snapshot Operation

◆ Snapshot Operation : Recover Snapshot(6)

- 주의사항 : 운영서비스 중지가 필요함. 즉 Downtime 필요함



- Compare file content 선택
 - Uncheck
 - Recover to spectied path
- Compare file property
 - 파일 속성 비교
- Compare file content
 - 파일 내용 비교
- Sync the whole file
 - 전체 파일 복구
- Bring in after finishing recovery
 - 복구 후에 Group Bring in
- Recover to specified path
 - 복구할 위치를 지정

- Recover Mode를 선택 후 Finish 선택
- 기존의 Data는 모두 삭제된다는 경고 메세지

7. Windows Update 절차

7. Windows Update 절차

◆ Windows Update 절차(Downtime 최소화 작업)

- Downtime 최소화 작업
 - 1) Active - Standby node 중 Standby node로부터 Windows Update 진행
 - 2) Standby node의 Windows Update 후 재시작
 - 3) Standby node의 재시작 후에 복제 정상 완료 확인
 - 4) Group Failover / Takeover 진행
 - 5) Standby node에서 서비스 정상 확인
 - 6) Active node의 Windows Update 후 재시작
 - 7) Active node의 재시작 후에 복제 정상 완료 확인
 - 8) 운영서비스는 Standby node에서 계속 운영
- Active node로 운영서비스 다시 절체
 - 1) Active - Standby node간의 Performance가 다를 때 사용
 - 2) 9)번까지 동일하게 진행
 - 3) Group Failover or Takeover 진행
 - 4) Active node에서 운영서비스 정상 확인 후 계속 운영

8. MSSQL or Oracle Patch 절차

8. MSSQL or Oracle Patch 절차

◆ 주의사항

- MSSQL or Oracle Patch 작업은 반드시 Downtime(서비스 중지)이 필요함
- MSSQL or Oracle Data File도 변경되기 때문이며, SQL or Oracle Service의 재시작 필요

◆ MSSQL or Oracle Patch 절차

- Downtime 최소화 작업
 - 1) Group Bring Out으로 운영서비스 중지
 - 2) Active - Standby node에 Service Pack or Patch 실행(동시에 진행)
 - 3) Service Pack or Patch 완료 후 Windows Service로 이동
 - 4) SQL or Oracle Service가 시작되어 있으므로 해당 서비스를 관리자가 중지함
 - 5) Group Bring In으로 운영서비스 시작 및 정상 확인
 - 6) Data의 verify가 완료되면, Group Failover or Takeover로 수동 절체함
 - 7) 운영서비스 정상 확인

9. 장애상황 대처방안

- Server 장애
- NIC 장애
- 운영서비스 장애
- OS Hangup으로 인한 장애

8. 장애상황 대처방안

◆ Server 장애

- Active node 장애
 - OS의 예기치 않은 종료 및 사용자의 실수로 인한 서비스 Shutdown / Restart가 발생하면 Standby node로 Failover 진행됨
- Standby node 장애
 - OS의 예기치 않은 종료 및 사용자의 실수로 인한 서버 Shutdown / Restart가 발생하면 Standby node이기 때문에 운영서비스에는 영향을 미치지 않음

◆ NIC(Network Interface Card) 장애

- Public 장애
 - Active node에서 Public Network 장애는 Standby node로 Failover 진행됨
 - Standby node에서 Public Network 장애는 운영서비스에 영향을 미치지 않음
- Heartbeat 장애
 - Active - Standby node에서 Heartbeat Network 장애는 운영서비스에 영향을 미치지 않음
- Mirror 장애
 - Active - Standby node에서 Mirror Network 장애는 운영서비스에 영향을 미치지 않음
 - 복제가 원활하게 진행되지 않기 때문에 조속한 복구가 필요함

8. 장애상황 대처방안

◆ 운영서비스 장애

- 운영서비스 즉, Oracle, SQL Service와 같은 운영에 필요한 서비스 장애인 경우 reset timeout까지 지정된 횟수를 재시작하며, 그 횟수를 초과하면 Failover 진행함
(Default setting : 3600s까지 3회 재시작, 변경 가능)

◆ OS Hang up으로 인한 장애

- RoseMirrorHA 및 기타 운영서비스(SQL or Oracle)도 OS를 기반으로 동작하는 Application이기 때문에 OS Hang up이 발생하면 정상적인 서비스가 어려움
- 만약 OS Hang up이 발생하면 운영서비스(Active node)를 강제종료함으로써, Failover를 진행할 수 있도록 하여 운영서비스를 정상화해야 함
- Standby node도 마찬가지로 OS Hang up이 발생할 경우, Active node의 장애가 발생하면 Failover를 진행할 수 없기 때문에 강제로 재시작하여, 정상적으로 대기할 수 있도록 조치를 취해야 함

10. Appendix

10. Appendix

◆ Replace IP 사용시 주의사항

- Replace IP를 사용하면 운영서버의 Real IP가 Virtual IP로 변경된다.
- 이 때에는 Login IP를 Real IP가 아닌 Heartbeat IP로 변경해주면 된다.
- 원격으로 운영서버를 접근할 경우, Virtual IP로 접근할 수 있다.

The screenshot displays the RoseMirrorHA Control Center interface for Windows. The main window shows two nodes: ROSEMIRROR01 and ROSEMIRROR02. ROSEMIRROR02 is selected, showing its properties in a table:

Property	Value
Data	ORA_Group
Connection	NA
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	NA
Mirror	NA

Below the main window, a log shows various system events, including a successful snapshot creation and a heartbeat timeout warning.

Overlaid on the bottom right is a Windows command prompt window titled "관리자: C:\Windows\system32\cmd.exe". It displays the IP configuration for three network adapters:

```
이더넷 어댑터 public:
연결별 DNS 접미사. . . . . :
IPv4 주소 . . . . . : 192.168.5.153
서브넷 마스크 . . . . . : 255.255.255.0
기본 게이트웨이 . . . . . : 192.168.5.1

이더넷 어댑터 heartbeat:
연결별 DNS 접미사. . . . . :
IPv4 주소 . . . . . : 10.10.50.151
서브넷 마스크 . . . . . : 255.255.255.0
기본 게이트웨이 . . . . . :

이더넷 어댑터 mirror:
연결별 DNS 접미사. . . . . :
IPv4 주소 . . . . . : 20.20.50.151
서브넷 마스크 . . . . . : 255.255.255.0
기본 게이트웨이 . . . . . :

터널 어댑터 isatap.{393CD8DA-F8E0-4291-BED4-1B94A9370A3B}:
-- More --
```

10. Appendix

◆ Replace IP 사용시 주의사항

- Heartbeat IP로 RoseMirrorHA Console에 접근

RoseMirrorHA Control Center - for Windows [rosemirror02]

System Server Private Net Data Application Service View Tools Help

Property Value

Data	ORA_Group
Connection	OK
Data Rule Set	ORA_Group
Source IP	20.20.50.151
Target IP	20.20.50.152
Speed (bytes...	33,064
Mirror	Idle
Left Bytes	0
Left Time	0s

Communication

Host: ROSEMIRROR01

IP List: 10.10.50.151 heartbeat

Port: 7330

OK Cancel

2014-07-08 16:18:34 | Login server successfully [ROSEMIRROR01 : ADMIN, GUI]

2014-07-08 16:11:26 | Create snapshot for the connection (ORA_Group) successfully, comment: (Vol:CD Type:Schedule). [ROSEMIRROR02]

2014-07-08 16:11:17 | Receive the signal of finishing synchronization in job (ORA_Group). [ROSEMIRROR02]

2014-07-08 16:11:03 | Failed to log in server [Unknow Error:16385] [ROSEMIRROR01, GUI]

2014-07-08 16:10:48 | The heartbeat of the private net (nah#2) is timeout over (15) seconds [ROSEMIRROR02]

10. Appendix

◆ 보안솔루션 설정

- Open RoseMirrorHA Port : TCP/UDP 7330, 3000, 3001, 3002, 7320
- RoseMirrorHA 설치 경로 등록 : C:\Program Files\MirrorHA
- RoseMirrorHA Process 등록 : MirrorHAService.exe, MirrorService.exe, MirrorMonitor.exe

◆ Windows Firewall 설정방법(Inbound port and process 설정)

- Open RoseMirrorHA Port : TCP/UDP 7330, 3000, 3001, 3002, 7320
- RoseMirrorHA Process 등록 : MirrorHAService.exe, MirrorService.exe, MirrorMonitor.exe

◆ RoseMirrorHA Service 중지 시 주의사항

- 관리자가 RoseMirrorHA의 서비스를 중지하면 운영서비스도 같이 중지됨으로, 관련 작업이 필요할 경우 엔지니어에게 요청

감사합니다.

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